

Module 2: Modulation



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M2L7: Objectives of Today Lecture

- Motivation to Modulation

Baseband Transmission

Requirements:

- Availability of channel bandwidth at-least equal to the BW of information signal.
- That BW should be exclusively used for that information signal.

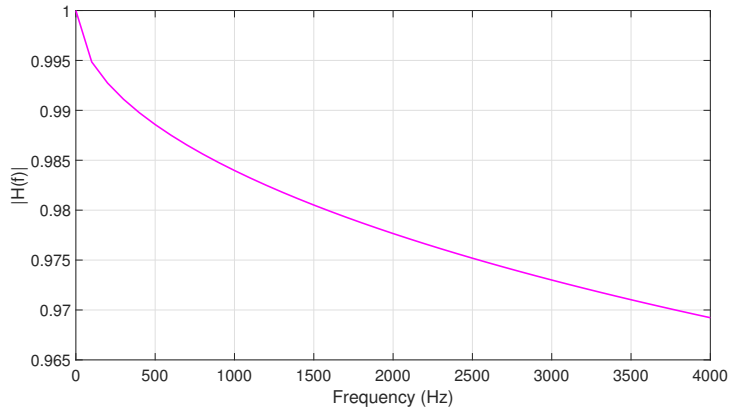
million?

Multiple Transmission in Baseband

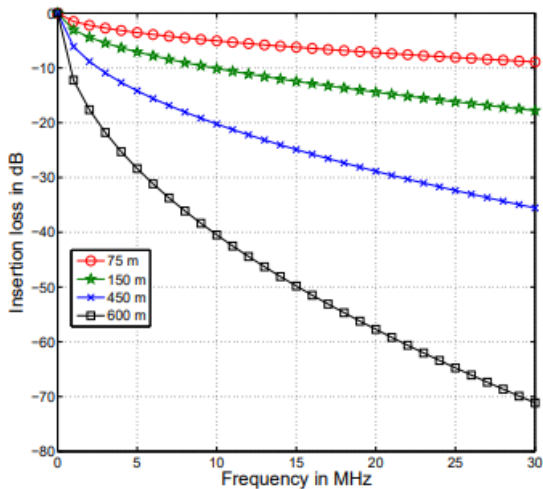
Is it possible to have baseband transmission for multiple information signal having the same bandwidth?

No

Telephone Channel: Gain versus Distance and Frequency



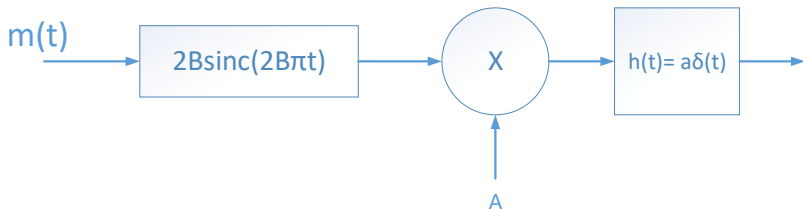
Telephone Channel: Gain versus Distance and Frequency



Transmission of Information Signal: Baseband

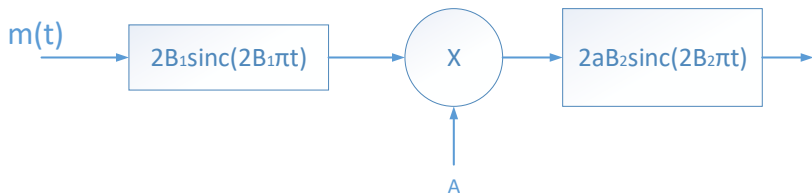
Baseband: Transmission of information signal without changing the operating frequency/center frequency

Block Diagram of Transmitter



Notation A will be changed

Block Diagram of Transmitter



$$B_2 > B_1$$

Notation A will be changed

Baseband to Passband Transmission

If baseband channel is not available, use passband transmission

- Offers flexibility in the utilization of vast EM spectrum
- Uses any carrier frequency depending on application/requirement
- Signal processing point view: Frequency translation from a low frequency to a higher carrier frequency

M2L8: Objectives of Today Lecture

- Motivation to Modulation Contd.

Transmit Power: Example

A signal $m(t) = \cos 2\pi f_m t$ is transmitted over channel $h(t) = a\delta(t)$. Design the transmitter to compensate power loss due to the channel when (i) $a = 1$ (ii) $a = 0.1$ (iii) $a = 10^{-6}$

Transmitter and Receiver Design

Problem: Revisit

Wire line System
Wireless System

M2L9: Objectives of Today Lecture

- Motivation to Modulation: Contd again

Signals: Bandpass and Lowpass

Lowpass Signal: $m(t)$ with frequency spectrum $M(f)$

Bandpass Signal: $x(t) = m(t) \cos 2\pi f_c t$

FT: $X(f) = M(f) * \frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]$

FT: $X(f) = \frac{1}{2}[M(f - f_c) + M(f + f_c)]$

Plot $|X(f)|$ with few different m_t types

Systems: Bandpass and Lowpass

Lowpass Filter: $g(t)$ with frequency spectrum $G(f)$

Bandpass Filter: $g(t) \cos 2\pi f_c t$

FT: $G(f) * \frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]$

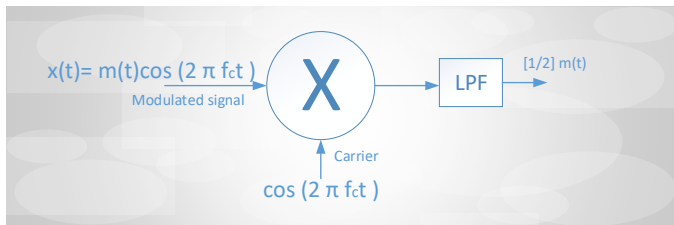
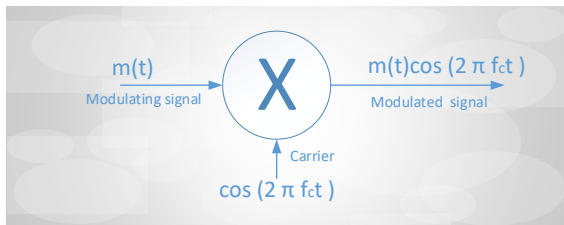
FT: $\frac{1}{2}[G(f - f_c) + G(f + f_c)]$

Plot: Baseband and Passband

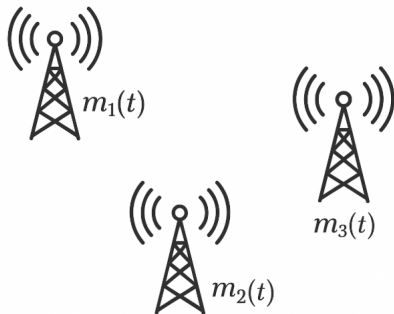
Baseband: $2B \text{sinc}(2B\pi t)$

Passband: $2B \text{sinc}(2B\pi t) \cos 2\pi f_c t$

Transmitter and Receiver: Single Station

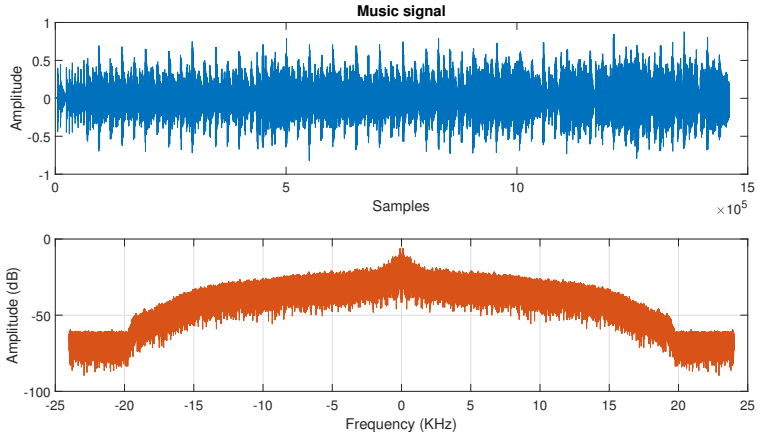


Transmitter and Receiver: Multiple Stations



- The frequency content of each signal is the same.
- Design transmitter and receiver.

Spectral Content of Information Signal: Music Signal



Sampling frequency 48KHz.

Spectral Content of Information Signal: Digital Signal

$$\Pi(t/T)$$
$$2B \text{sinc}(2B\pi t)$$

Spectral Content of Information Signal

In general: Information Signal are Low Pass Signal with non-zero BW

- BW of voice signal: 4 KHz
- BW of music signal: 24 KHz
- BW of TV signal: 4.2 MHz/5.5MHz

Other Issues with Baseband Wireless Transmission

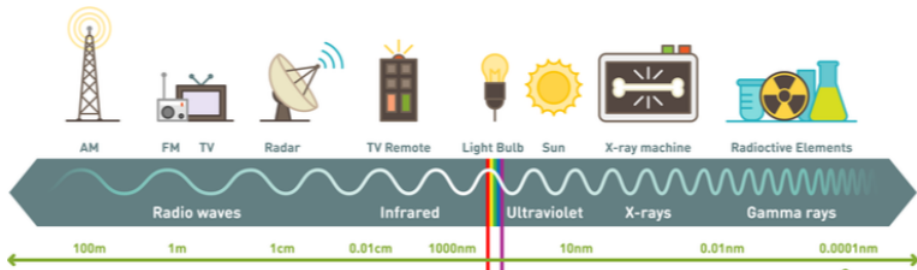
- Length of radiating antenna $\approx \lambda$
- Transmit power $\propto \lambda$
- Channel Noise and Interference: In the EM spectrum, there can be frequency bands which can offer lesser channel noise and interference

Electromagnetic Spectrum

S.No	Frequency Band	Wavelength	Applications
1.	30 Hz – 300 Hz. Extremely low frequencies(ELF)	10^4 km to 10^3 km	Power transmission
2.	300 Hz – 3 kHz. Voice frequencies (VF)	10^3 km to 100 km	Audio applications
3.	3 kHz – 30 kHz. Very low frequencies (VLF)	100 km to 10 km	Submarine communications. Navy, Military communications
4.	30 kHz – 300 kHz . Low frequencies (LF)	10 km to 1 km. Long waves.	Aeronautical and marine navigation, these frequencies act as sub carriers.
5.	300 kHz – 30 MHz. Medium frequencies (MF)	1 km to 100 m. Medium waves	AM radio broadcast, Marine and aeronautical communications
6.	3 MHz – 30 MHz. High Frequencies (HF)	100 m TO 10 m. Short waves	Shortwave transmission, Amateur and CB communication.
7.	30 MHz – 300 MHz. Very high frequencies (VHF)	10 m to 1 m	T.V. broadcasting. F.M. broadcasting

Electromagnetic Spectrum

Electromagnetic Spectrum



A typical Radio Receiver



Definition of Modulation

- Let $m(t)$: message signal of bandwidth B .
- Let $A \cos(2\pi f_c t + \phi(t))$: Carrier signal
- There are three variables in the carrier signal: amplitude A , frequency f , phase ϕ
- *Modulation is defined as the process by which some characteristics of a carrier signal is varied in accordance with a modulating wave (or message signal or information-bearing signal).*
- The resultant signal is called as : Modulated wave.

AM Radio Band

505 KHz to 1605 KHz

- Can we transmit voice signal in the AM band?
How many?
- Can we transmit music signal in the AM band?
How many?
- Can we transmit video signal in the AM band?
How many?

FM Radio Band

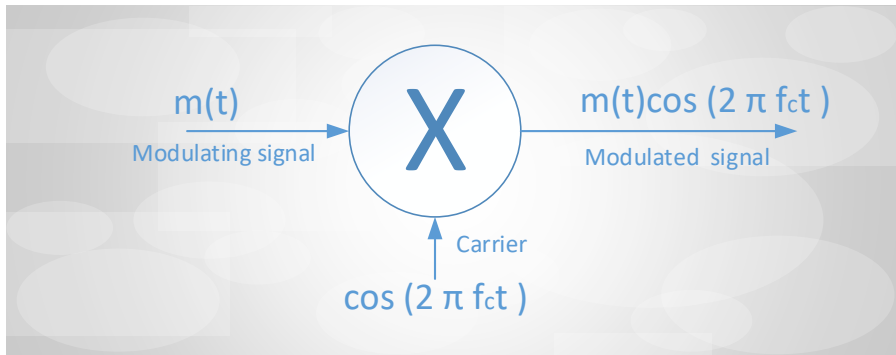
88 MHz to 108 MHz

- Can we transmit voice signal in the FM band?
How many?
- Can we transmit music signal in the FM band?
How many?
- Can we transmit video signal in the FM band?
How many?

Analog Modulation

- Amplitude Modulation $A \propto m(t)$
- Angle Modulation $(2\pi f_c t + \phi) \propto m(t)$
- Phase Modulation $\phi \propto m(t)$
- Frequency Modulation $f_c \propto m(t)$

Block Diagram of Transmitter



Demodulation

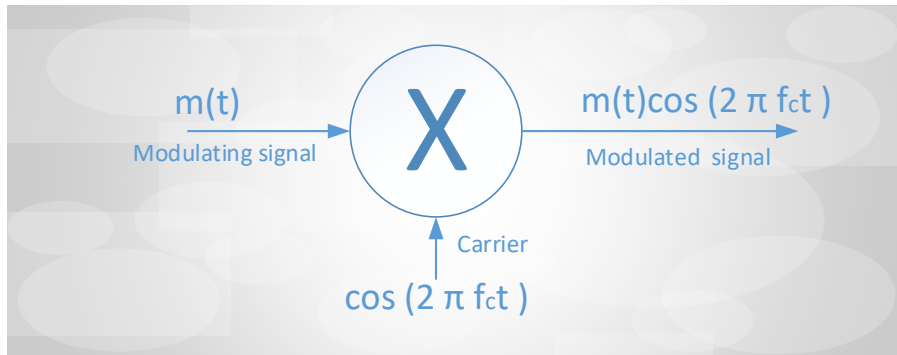
Demodulation is the process by which the original modulating (or message signal or information-bearing signal) is recovered from the modulated wave.

A system that performs both modulation and demodulation: modem

Amplitude Modulation

- Message signal $m(t)$
- Carrier Signal: $c(t) = A \cos(2\pi f_c t)$
- Amplitude Modulation $A \propto m(t)$
- Modulated signal $x(t) = m(t) \cos 2\pi f_c t$

Block Diagram of Transmitter



Waveform

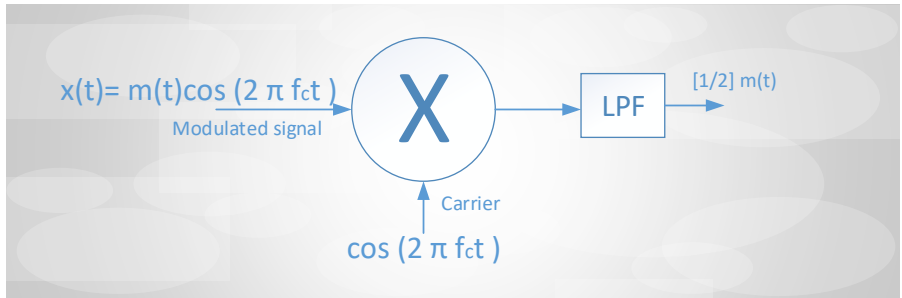
- Time-Domain
- Frequency-Domain
- Bandwidth

Type of AM? DSB-SC

Power

- $x(t) = m(t) \cos(2\pi f_c t)$
- ACF: $R_x(\tau) = \frac{1}{2} R_m(\tau) \cos 2\pi f_c \tau$
- Power = $R_x(0) = \frac{1}{2} R_0(\tau) = \frac{1}{2} P_m$
- $S_x(f) = \frac{1}{4} (S_m(f - f_c) + S_m(f + f_c))$.

Demodulation of DSB-SC: Synchronous



M2L10: Objectives of Today Lecture

- AM: DSB-SC with Carrier
- Modulation Index
- Power Efficiency

How to Study Analog Modulation

- Time-domain: Waveform of the modulated signal
- Frequency domain: To get the bandwidth of the modulated signal
- Generation of modulated signal: Circuits to generate modulated signals
- Recovery of message signals: Circuits of demodulation
- Comparison of different modulation schemes with respect to: bandwidth, power, and transceiver complexity.

Synchronous Demodulation: A Practical Problem

A DSB-SC modulated signal $x(t) = Am(t) \cos 2\pi f_c t$ is mixed with a local carrier $\cos(2\pi f_c t + \theta)$ and the output is passed through a LPF with a bandwidth equal to the bandwidth of the message signal $m(t)$. Plot the output power versus θ for $0 \leq \theta \leq \pi$.

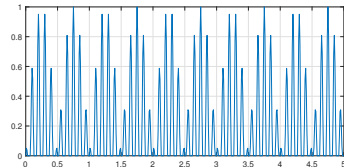
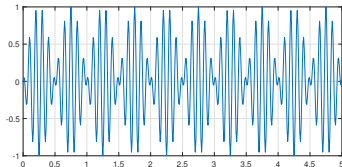
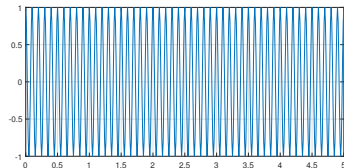
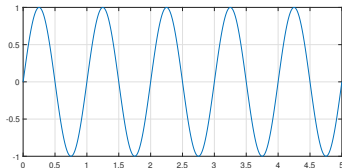
Solution

- Modulated signal $x(t) = m(t) \cos 2\pi f_c t$
- Demodulation process
- Mixing: $y_1(t) = x(t) \cos(2\pi f_c t + \theta)$
- $y_1(t) = m(t) \cos 2\pi f_c t \cos(2\pi f_c t + \theta)$
- $\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$

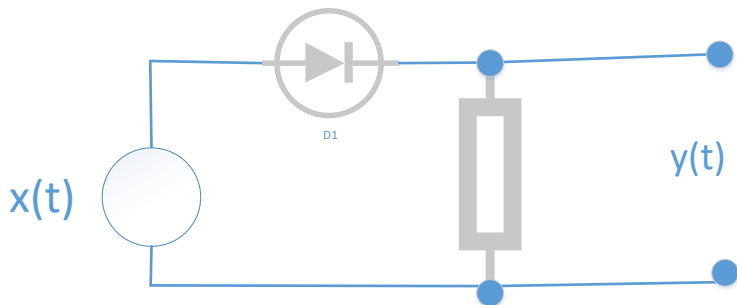
Solution

- $y_1(t) = \frac{1}{2}m(t)[\cos 4\pi f_c t + \cos(\theta)]$
- Apply LPF filter at the center frequency of $M(f)$ and BW of $m(t)$
- $y_{out}(t) = \frac{1}{2}m(t) \cos(\theta)$
- $P_{out} = \frac{1}{4} \cos^2(\theta) P_m$
- $P_{out} = \frac{1}{4} P_m$ at $\theta = 0$;
- $P_{out} = 0$ (quadrature effect) at $\theta = \pi/2$.
- $P_{out} = \frac{1}{4} P_m$ at $\theta = \pi$.

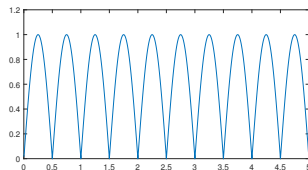
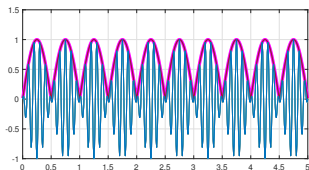
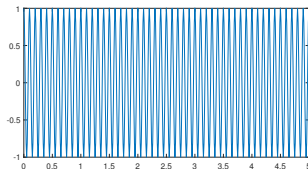
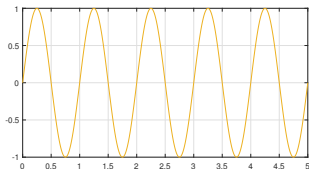
Demodulation: Envelope Detection



Envelop Detection

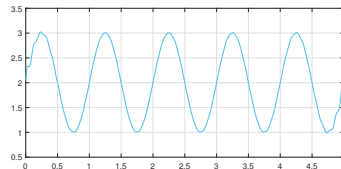
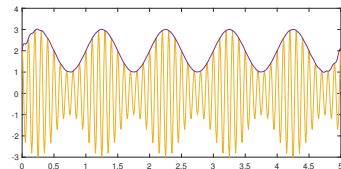
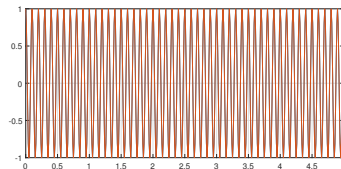
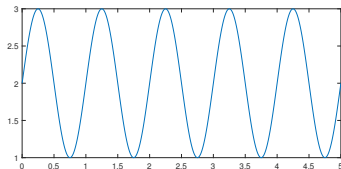


Envelop Detection



Envelope Detection

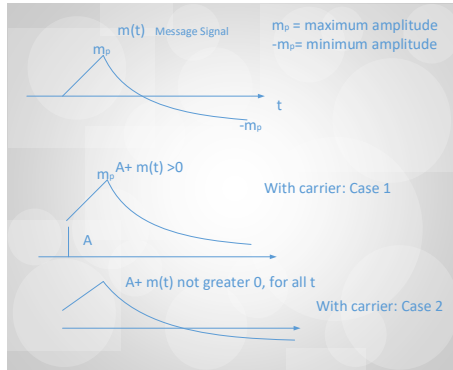
Signal must have a single polarity.



Amplitude Modulation

- AM: DSB-SC with Carrier
- $x(t) = A \cos(2\pi f_c t) + m(t) \cos(2\pi f_c t)$
- $x(t) = (A + m(t)) \cos(2\pi f_c t)$
- Modulation index $\mu = \frac{m_p}{A}$, where m_p is the maximum amplitude of $m(t)$.
- What is $\mu = 1$, $\mu = 0$ and $\mu = \infty$?
- What is advantage and disadvantage of adding a carrier?

Amplitude Modulation: Time-Domain



- Draw the AM waveform

AM: Frequency-Domain

- $x(t) = [A + m(t)] \cos(2\pi f_c t)$
- Draw the frequency spectrum
- $X(f) =$
 $\frac{1}{2}[M(f + f_c) + M(f - f_c)] + \frac{A}{2}[\delta(f + f_c) + \delta(f - f_c)]$

Amplitude Modulation: Power Computation

- Carrier: $A \cos(2\pi f_c t)$
- Sidebands: $m(t) \cos(2\pi f_c t)$
- Carrier power: $P_c = \frac{A^2}{2}$.
- Sidebands power $P_s = \frac{1}{2}P_m$, where P_m is the average power of $m(t)$.
- Power efficiency: $\eta = \frac{P_s}{P_s + P_c} = \frac{P_m}{A^2 + P_m} 100\%$

Power Efficiency for Tone-modulation

- Carrier: $A \cos(2\pi f_c t)$
- Message $m(t) = A_m \cos(2\pi f_m t)$
- Power efficiency: $\eta = \frac{\mu^2}{\mu^2 + 2} \times 100\%$

Bandwidth of DSB-SC versus AM

Both $2B$

CW Problem

- Determine modulation index of AM modulated signal
- $x(t) = \frac{3}{2} \cos(190 \times 10^3 \pi t) + \frac{3}{2} \cos(210 \times 10^3 \pi t) + 5 \cos(200 \times 10^3 \pi t)$

Solution

- Use identity
$$\cos(A) + \cos(B) = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$
- $x(t) = [5 + 3 \cos(10 \times 10^3 \pi t)] \cos(200 \times 10^3 \pi t)$
- $\mu = 0.6$: modulation index.
- $m(t) = 3 \cos(10 \times 10^3 \pi t)$
- Carrier signal: $m(t) = 5 \cos(200 \times 10^3 \pi t)$

M2L11: Objectives of Today Lecture

- Envelope Detection
- Some Important Problems

Envelope Detection: Signal Processing

Problem Statement: How to extract amplitude A from $A \cos(2\pi f_c t)$

Hint: Use Hilbert Transform

Envelope Detection: Signal Processing

Problem Statement: How to extract amplitude $m(t)$ from $m(t) \cos(2\pi f_c t)$

Hint: Use Hilbert Transform

Hilbert Transform of Band-pass Signal

Let $c(t)$ be a bandpass signal with spectrum $C(f)$ concentrated around $\pm f_c$, and let $m(t)$ be a lowpass signal with Fourier transform $M(f)$ such that their spectra do not overlap.

Define

$$x(t) = m(t)c(t).$$

Then the Hilbert transform of $x(t)$ is given by

$$x_h(t) = m(t)c_h(t),$$

where $c_h(t)$ denotes the Hilbert transform of $c(t)$.

Hilbert Transform of $x(t) = m(t)c(t)$

- Express $C(f) = C_+(f) + C_-(f)$.
- FT of $m(t)c(t)$

$$X(f) = M(f) * C(f),$$

- Using above:

$$X(f) = M(f) * [C_+(f) + C_-(f)].$$

- $X_+(f) = M(f) * C_+(f)$; $X_-(f) = M(f) * C_-(f)$
- Hilbert transform in frequency domain:

$$\mathcal{F}\{x_h(t)\} = -j \operatorname{sgn}(f) X(f) = -jX_+(f) + jX_-(f).$$

$$\mathcal{F}\{x_h(t)\} = M(f) * [-jC_+(f) + jC_-(f)].$$

$$\mathcal{F}\{x_h(t)\} = M(f) * \mathcal{F}\{c_h(t)\} \Rightarrow x_h(t) = m(t)c_h(t).$$

CW Problem

Find the Hilbert Transform of the following signal:

① $g(t) = \text{sinc}(\pi t)$

Solution 1

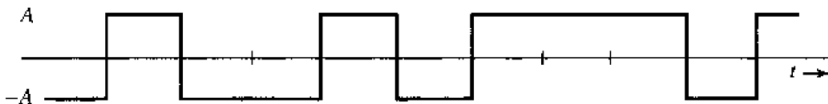
- Expressing $g(t) = \text{sinc}(\pi t) = \frac{\sin \pi t}{\pi t} = \frac{2 \sin \pi t/2 \cos \pi t/2}{\pi t} = \text{sinc}(\pi t/2) \cos(\pi t/2)$.
- Using Hilbert identity for product of low pass-high pass signal $H(m(t)c(t)) = m(t)H(c(t))$,
 $g_h(t) = \text{sinc}(\pi t/2) \sin(\pi t/2)$.

Envelope Detection: Signal Processing

Problem Statement: How to extract amplitude $m(t)$ from $[A + m(t)] \cos(2\pi f_c t)$

CW Problem

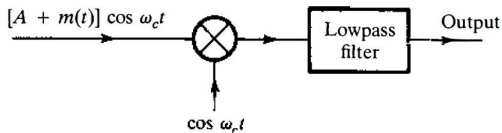
- 1 Analyze the power efficiency of a AM signal with $\mu = 1$ for a message signal as given in the figure:



CW Problem

4.3-1 Figure P4.3-1 shows a scheme for coherent (synchronous) demodulation. Show that this scheme can demodulate the AM signal $[A + m(t)] \cos(2\pi f_c t)$ regardless of the value of A

Figure P.4.3-1



CW Problem

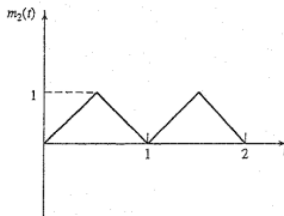
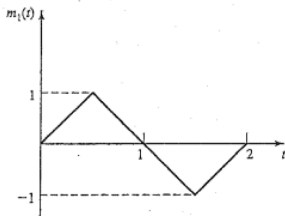
In a DSB-SC system the carrier is $A \cos 2\pi f_c t$ and the message signal is given by $m(t) = \text{sinc}(\pi t) + \text{sinc}^2(\pi t)$. Find the bandwidth of the modulated signal.

Solution

- $m_1(t) = \text{sinc}(\pi t)$
- $M_1(f) = \Pi(f)$
- $m_2(t) = \text{sinc}^2(\pi t)$
- FT pair $B\text{sinc}^2(\pi Bt) \rightarrow \Delta(\frac{f}{2B})$
- $M_2(f) = \Delta(f/2)$
- Plot baseband and passband spectrum.
- BW baseband: 1 Hz
- BW passband: 2 Hz

Problem

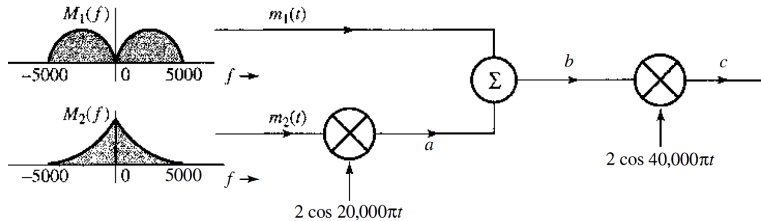
Sketch modulated signals $x_1(t) = m_1(t) \cos 2\pi f_c t$ and $x_2(t) = m_2(t) \cos 2\pi f_c t$ for a duration of 2 seconds. Find the average power of $x_1(t)$ and $x_2(t)$. Find additional transmit power added for $m_1(t)$ to enable envelope detection. Take $f_c = 5$ Hz.



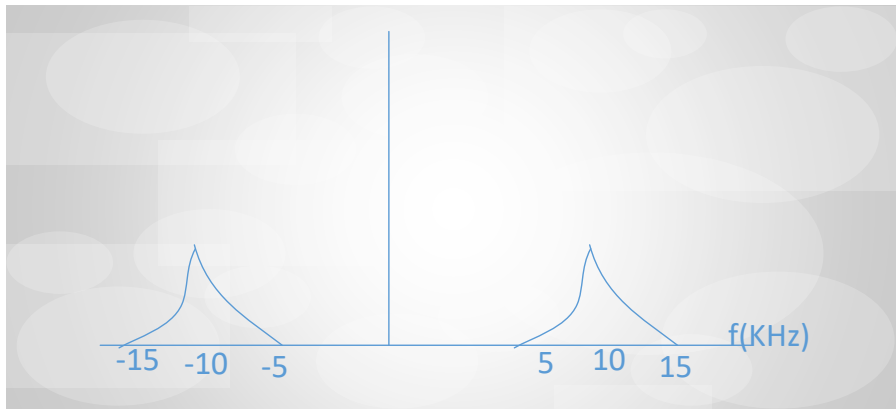
TB problem

Two signals $m_1(t)$ and $m_2(t)$, both band-limited to 5000 Hz, are to be transmitted simultaneously over a channel by the multiplexing scheme shown in Fig. 1. The signal at point b is the multiplexed signal, which now modulates a carrier of frequency 20,000 Hz. The modulated signal at point c is transmitted over a channel. (a) Sketch signal spectra at points a, b, and c. (b) What must be the bandwidth of the channel? (c) Design a receiver to recover signals $m_1(t)$ and $m_2(t)$ from the modulated signal at point c.

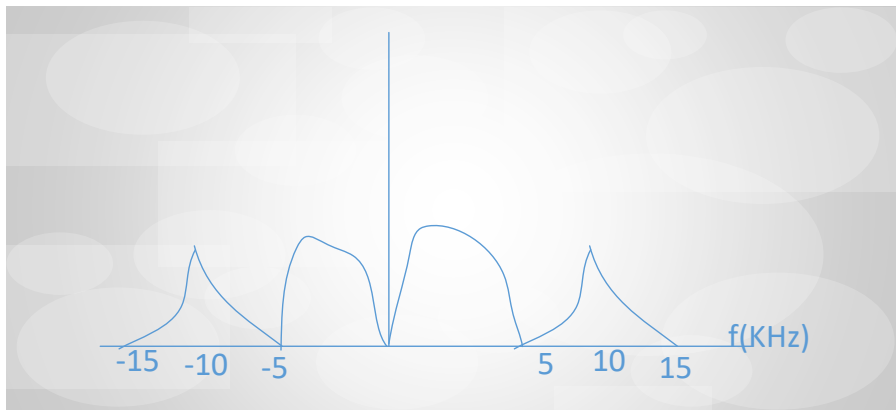
TB Problem



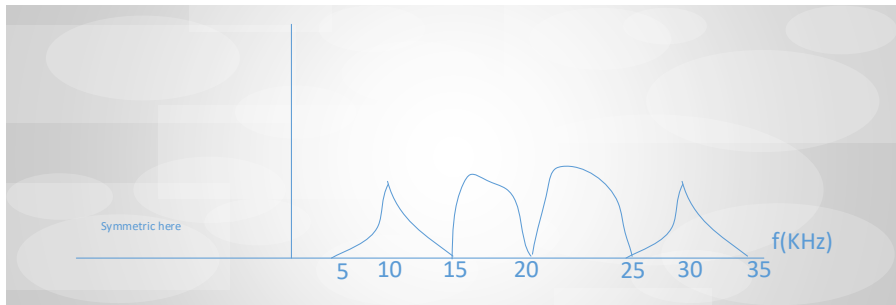
Solution a: Sketch at a



Solution a: Sketch at b



Solution a: Sketch at c



Solution b:

$$\text{BW: } 35-5=30 \text{ KHz}$$

Solution c:

Signal at c: $2m_1(t) \cos(40,000\pi t) + 2m_2(t) \cos(20,000\pi t) \cos(40,000\pi t)$.

Multiply with synchronous carriers and apply LPF to get $m_1(t)$ and $m_2(t)$.

M2L12: Objectives of Today Lecture

- Modulation Index
- Some Important Problems



Hilbert Transform of $x(t) = m(t)c(t)$: Revisit

- Express $C(f) = C_+(f) + C_-(f)$.
- FT of $m(t)c(t)$

$$X(f) = M(f) * C(f),$$

- Using above:

$$X(f) = M(f) * [C_+(f) + C_-(f)].$$

- $X_+(f) = M(f) * C_+(f)$; $X_-(f) = M(f) * C_-(f)$
- Hilbert transform in frequency domain:

$$\mathcal{F}\{x_h(t)\} = -j \operatorname{sgn}(f) X(f) = -jX_+(f) + jX_-(f).$$

$$\mathcal{F}\{x_h(t)\} = M(f) * [-jC_+(f) + jC_-(f)].$$

Modulation Index: Symmetric

Using $\mu = \frac{m_p}{A}$; Show that:

$$\textcircled{1} \quad \mu = \frac{x_{max} - x_{min}}{x_{max} + x_{min}}$$

Proof

- ① $x_{max} = A + m_p$
- ② $x_{max} = A - m_p$
- ③ Add to get $A = \frac{x_{max} + x_{min}}{2}$
- ④ Subtract $m_p = \frac{x_{max} - x_{min}}{2}$
- ⑤ $\mu = \frac{m_p}{A} = \frac{x_{max} - x_{min}}{x_{max} + x_{min}}$

Modulation Index: Asymmetric

Prove that: $\mu = \frac{m_{\max} - m_{\min}}{2A + m_{\max} + m_{\min}}$

Assume:

$$m_{\max}, m_{\min} \quad (m_{\max} \neq -m_{\min})$$

- $x(t) = (A + m(t)) \cos 2\pi f_c t$

- Define:

$$m_0 = \frac{m_{\max} + m_{\min}}{2}$$

- Define a symmetric signal:

$$\tilde{m}(t) = m(t) - m_0, \quad \tilde{m}_p = \frac{m_{\max} - m_{\min}}{2}$$

- Now $x(t) = (A + m_0 + m(t) - m_0) \cos 2\pi f_c t$

$$\mu = \frac{\frac{m_{\max} - m_{\min}}{2}}{A + \frac{m_{\max} + m_{\min}}{2}} = \frac{m_{\max} - m_{\min}}{2A + m_{\max} + m_{\min}}$$

General: Modulation Index

Show that in general:

$$\mu = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$$

- Rewrite the asymmetric case as:

$$\mu = \frac{m_{\max} + A - m_{\min} - A}{A + m_{\max} + A + m_{\min}}$$

- Using $m_{\max} + A = x_{\max}$ and $m_{\min} + A = x_{\min}$
- Thus,

$$\mu = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$$

Problem on AM with Asymmetric Message Signal

A message signal has:

$$m_{max} = 6 \text{ V}, \quad m_{min} = -2 \text{ V}$$

This message is amplitude modulated with a carrier of amplitude:

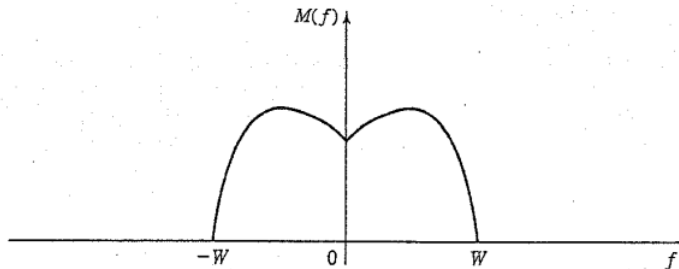
$$A = 10 \text{ V}$$

Find the modulation index μ .

Answer 1/3

CW Problem

A signal $x(t) = m(t) + \cos 2\pi f_c t$ is applied to a non-linear system $y(t) = x(t) + \frac{1}{2}x^2(t)$. The message $m(t)$ has a bandwidth W with frequency spectrum as shown in the figure. Determine the specification of a band-pass filter applied to the signal $y(t)$ such that AM signal is transmitted.



Solution

Generation of AM signal using non-linear method:

$$y(t) = x(t) + \frac{1}{2}x^2(t)$$

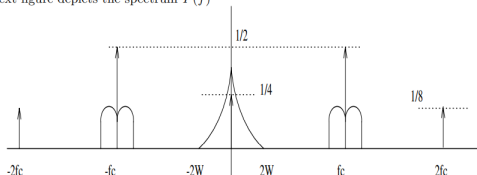
43

$$\begin{aligned} &= m(t) + \cos(2\pi f_c t) + \frac{1}{2} \left(m^2(t) + \cos^2(2\pi f_c t) + 2m(t) \cos(2\pi f_c t) \right) \\ &= m(t) + \cos(2\pi f_c t) + \frac{1}{2}m^2(t) + \frac{1}{4} + \frac{1}{4} \cos(2\pi 2f_c t) + m(t) \cos(2\pi f_c t) \end{aligned}$$

Taking the Fourier transform of the previous, we obtain

$$\begin{aligned} Y(f) &= M(f) + \frac{1}{2}M(f) \star M(f) + \frac{1}{2}(M(f - f_c) + M(f + f_c)) \\ &\quad + \frac{1}{4}\delta(f) + \frac{1}{2}(\delta(f - f_c) + \delta(f + f_c)) + \frac{1}{8}(\delta(f - 2f_c) + \delta(f + 2f_c)) \end{aligned}$$

The next figure depicts the spectrum $Y(f)$



Solution: BPF

Note that the spectrum at f_c is from $f_c - W$ to $f_c + W$. To transmit an AM signal, a bandpass filter of center frequency f_c and width $2W$ should be applied. Ignore the gain of the BPF.

CW Problem

3.18 The message signal $m(t)$ whose spectrum is shown in Figure P-3.18 is passed through the system shown in the same figure.

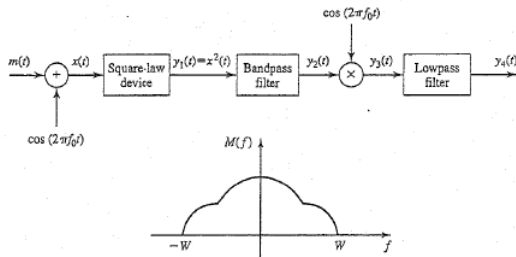


Figure P-3.18

The specification of BPF is $2W$ bandwidth centered at f_0 and LPF has a BW of W . Plot the spectra and find the bandwidth of $y_1(t)$, $y_2(t)$, $y_3(t)$, $y_4(t)$.

Solution

$$x(t) = m(t) + \cos(2\pi f_0 t).$$

$$X(f) = M(f) + \frac{1}{2}(\delta(f - f_0) + \delta(f + f_0)), \quad B_x = f_0.$$

After Square Law Device:

$$y_1(t) = x^2(t) = m^2(t) + \frac{1}{2} + \frac{1}{2} \cos(2\pi \cdot 2f_0 t) + 2m(t) \cos(2\pi f_0 t).$$

Spectrum:

$$Y_1(f) = M(f) \star M(f) + \frac{1}{2}\delta(f) + \frac{1}{4}(\delta(f - 2f_0) + \delta(f + 2f_0)) + M(f - f_0) + M(f + f_0).$$

Bandwidth $2W$

Solution

After BPF:

$$Y_2(f) = M(f - f_0) + M(f + f_0), \quad W_2 = 2W.$$

Next

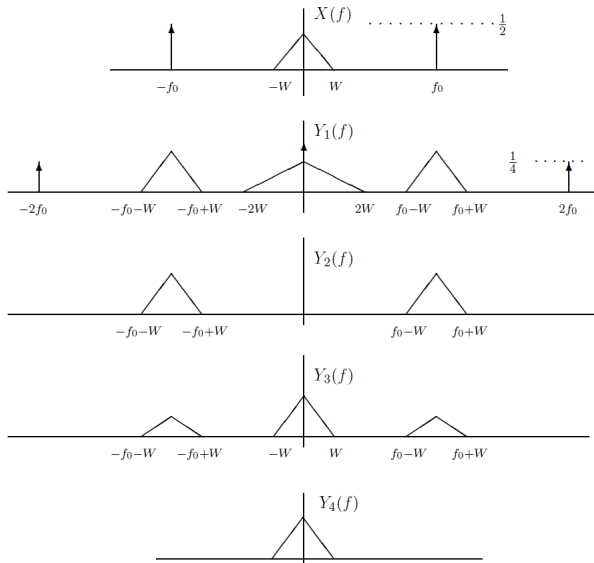
$$y_3(t) = 2m(t) \cos^2(2\pi f_0 t) = m(t) + m(t) \cos(2\pi f_0 t).$$

$$Y_3(f) = M(f) + \frac{1}{2}(M(f - f_0) + M(f + f_0)), \quad W_3 = f_0 + W.$$

After Lowpass filter:

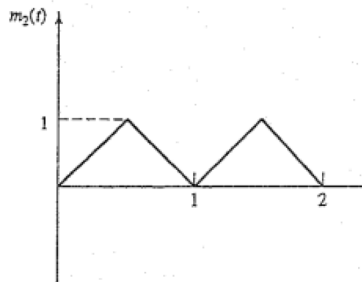
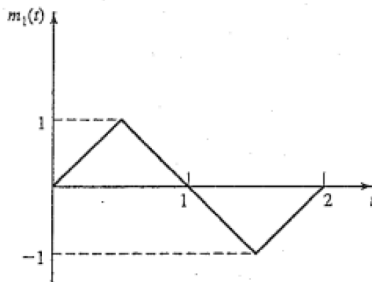
$$y_4(t) = m(t), \quad Y_4(f) = M(f), \quad W_4 = W.$$

Solution



CW Problem

Sketch modulated signals $x_1(t) = m_1(t) \cos 2\pi f_c t$ and $x_2(t) = m_2(t) \cos 2\pi f_c t$ for a carrier wave $\cos 2\pi f_c t$. Find the average power of $x_1(t)$ and $x_2(t)$. Find additional transmit power added by envelope detection. Take $f_c = 5$ Hz.

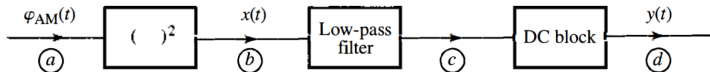


Solution

- $P_{m1} = P_{m2} = \frac{1}{1/2} \int_0^{1/2} (2t)^2 dt = \frac{1}{3},$
- $P_{x1} = P_{x2} = 1/6.$
- General AM signal is $(A + m(t)) \cos 2\pi f_c t.$
- Carrier to be added $\cos 2\pi f_c t$, i.e., $A = 1.$
- Additional power $= \frac{1}{2}.$

CW Problem

4.3-8 In the early days of radio, AM signals were demodulated by a crystal detector followed by a low-pass filter and a dc blocker, as shown in Fig. P4.3-8. Assume a crystal detector to be basically a squaring device. Determine the signals at points a , b , c , and d . Point out the distortion term in the output $y(t)$. Show that if $A \gg |m(t)|$, the distortion is small.



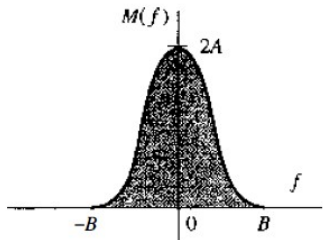
Signal at (c):
$$\frac{A^2 + 2Am(t) + m^2(t)}{2}$$

M2L13: Objectives of Today Lecture

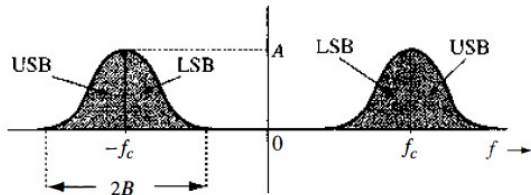
- Single Side Band (SSB) Modulation



Spectra of Modulating and Modulated Signals

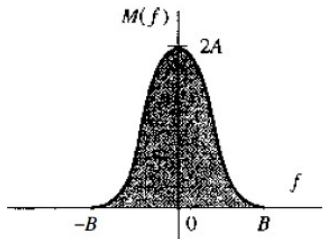


(a)

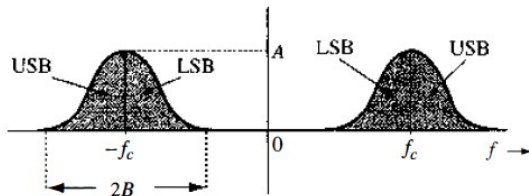


(b)

Spectra of Modulating and Modulated Signals

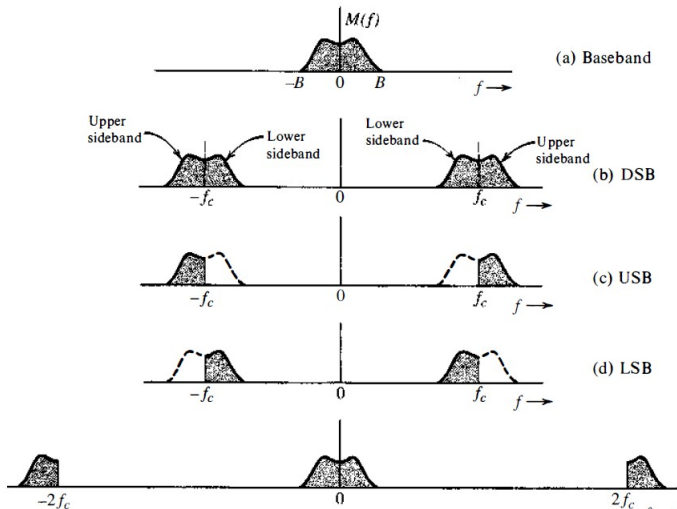


(a)

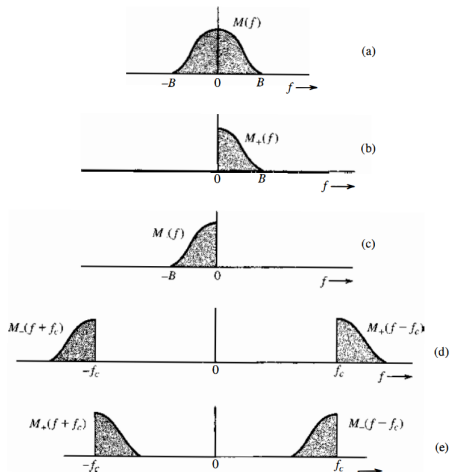


(b)

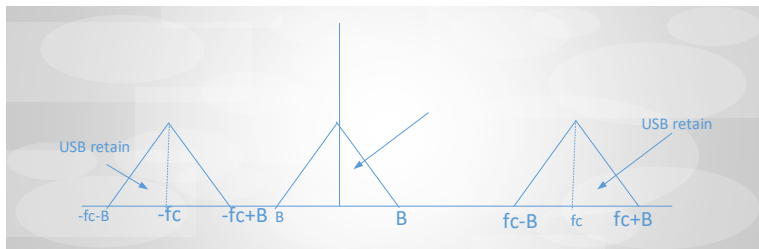
Spectra of Modulating and Modulated Signals



Spectra of Modulating and Modulated Signals



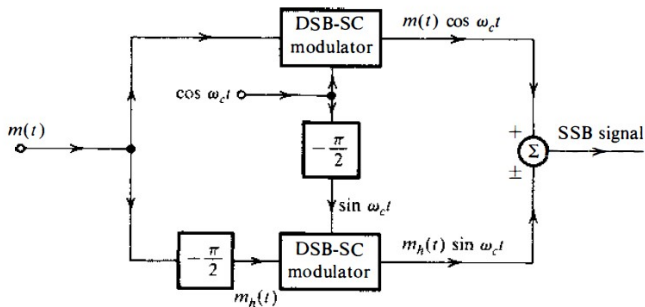
Spectra



Bandwidth Efficient AM: SSB

- $M_+(f) = M(f)\frac{1}{2}[1 + \text{sgn}(f)] = \frac{1}{2}[M(f) + jM_h(f)]$
- $M_-(f) = M(f)\frac{1}{2}[1 - \text{sgn}(f)] = \frac{1}{2}[M(f) - jM_h(f)]$
- USSB Signal $X(f) = M_+(f - f_c) + M_-(f + f_c)$
- $X(f) = \frac{1}{2}[M(f - f_c) + M(f + f_c)] - \frac{1}{2j}[M_h(f - f_c) - M_h(f + f_c)]$
- $x_{USSB}(t) = m(t) \cos(2\pi f_c t) - m_h(t) \sin(2\pi f_c t)$
- Similarly,
 $x_{LSSB}(t) = m(t) \cos(2\pi f_c t) + m_h(t) \sin(2\pi f_c t)$

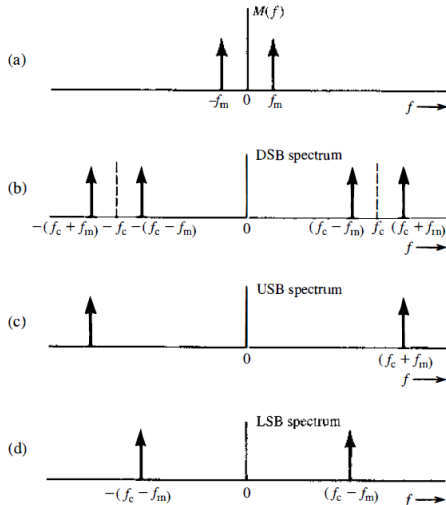
Block Diagram



SSB: Single-Tone Message

- $m(t) = \cos(2\pi f_m t)$, carrier $\cos(2\pi f_c t)$
- Write the transmission equation for upper and lower SSB signals.
- Directly:
$$x_{USSB}(t) = m(t) \cos(2\pi f_c t) - m_h(t) \sin(2\pi f_c t) = \cos(2\pi f_c + 2\pi f_m)t \text{ and}$$
$$x_{LSSB}(t) = m(t) \cos(2\pi f_c t) + m_h(t) \sin(2\pi f_c t) = \cos(2\pi f_c - 2\pi f_m)t$$
- $\cos A \cos B - \sin A \sin B = \cos(A + B)$
- $\cos A \cos B + \sin A \sin B = \cos(A - B)$

SSB: Single-Tone Message (Frequency Domain approach)



SSB: Power and Bandwidth

Half of the bandwidth than DSB-SC requires double of the transmit power.

Problem

A message signal has a bandwidth of 10 KHz, and its power content is 0.5 W. The carrier signal $A \cos(2\pi f_c t)$ has a power content of 200 W. Find the bandwidth and power content of the following modulated signals:

- 1 DSB-SC
- 2 AM Signal
- 3 SSB

Solution: DSB-SC

- 1 Bandwidth of DSB-SC = $2 \times 10 \text{ KHz} = 20 \text{ KHz}$.
- 2 DSB-SC signal $x(t) = m(t) \cos(2\pi f_c t)$.
- 3 The average power content $P_{DSB-SC} = (1/2)P_m$, where P_m is the average power of the message signal $m(t)$.
- 4 $P_{DSB-SC} = 0.5/2 = 0.25$.

Solution: AM

- 1 Bandwidth of AM = $2 \times 10 \text{ KHz} = 20 \text{ KHz}$.
- 2 AM signal $x(t) = (A + m(t)) \cos(2\pi f_c t)$.
- 3 The average power content
 $P_{AM} = (A^2/2) + P_m/2$, where P_m is the average power of the message signal $m(t)$.
- 4 $P_{AM} = 200.25$.

Solution: SSB

- 1 Bandwidth of SSB 10KHz.
- 2 SSB signal
$$x(t) = m(t) \cos(2\pi f_c t) \pm m_h(t) \sin(2\pi f_c t).$$
- 3 The average power content
$$P_{SSB} = (P_m/2) + (P_{mh}/2)$$
- 4 P_m is the average power of the message signal $m(t)$
- 5 P_{mh} is the average power of the message signal $m(t)$ after Hilbert transform.
- 6 $P_{SSB} = 0.5$.

CW Problem

Find the USSB and LSSB time domain for the modulating signal

① $m(t) = \text{sinc}(2\pi Bt)$

② $m(t) = \frac{B}{2} \text{sinc}^2(\pi Bt)$

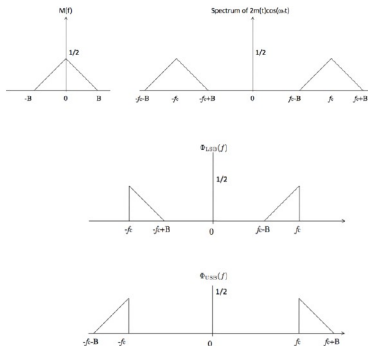
with $B = 2000$, $f_c = 10,000$, and $A_c = 1$.

Solution

First part: time-domain directly

For the second part:

(ii) Use frequency-domain approach: $M(f) = \frac{1}{2} \Delta(\frac{f}{2B})$



M2L14: Objectives of Today Lecture

- Single Side Band (SSB) Demodulation
- Circuits to generate AM Signals



SSB Modulated Signal

- $x_{USSB}(t) = m(t) \cos(2\pi f_c t) - m_h(t) \sin(2\pi f_c t)$
- $x_{LSSB}(t) = m(t) \cos(2\pi f_c t) + m_h(t) \sin(2\pi f_c t)$

SSB : Frequency-Domain Approach

Few General Signal Type.

CW Problem

A message signal $m(t) = 2\text{sinc}(2\pi t)$ modulates the carrier $\cos 20\pi t$. Find the time-domain USSB signal and frequency spectrum of the USSB signal.

SSB : CW Problem

$$m(t) = 2 \cos(2\pi \cdot 1 \text{ kt}) + \frac{1}{2} \cos(2\pi \cdot 2 \text{ kt}),$$

Carrier signal at $f_c = 100 \text{ kHz}$. Find time-domain USSB signal.

Solution

$$\hat{m}(t) = 2 \sin(2\pi \cdot 1 \text{ kt}) + \frac{1}{2} \sin(2\pi \cdot 2 \text{ kt}).$$

$$x_{SSB}(t) = m(t) \cos(2\pi f_c t) - \hat{m}(t) \sin(2\pi f_c t).$$

$$x_{SSB}(t) = 2 \cos(2\pi \cdot 101 \text{ kt}) + \frac{1}{2} \cos(2\pi \cdot 102 \text{ kt}).$$

SSB: De-modulator

- Coherent Demodulation: Multiply the received SSB signal by $\cos 2\pi f_c t$ plus LPF.
- Envelop Detector: Transmit $x_{ssb}(t) + A \cos 2\pi f_c t$
- Show that envelop of signal
$$E(t) = \sqrt{(A + m(t))^2 + m_h^2(t)}$$
- $E(t) \approx A + m(t)$, if $A \gg |m(t)|$
- SSB requires more carrier power to be added for envelop detection.

TB problem

A USB signal is generated by using the phase shift method. If the input to this system is $m_h(t)$ instead of $m(t)$, what will be the output? Is this signal still an SSB signal with bandwidth equal to that of $m(t)$? Can this signal be demodulated [to get back $m(t)$]? If so, how?

Solution

- USB
- $y(t) = m_h(t) \cos 2\pi f_c t - m_{hh}(t) \sin 2\pi f_c t$
- $y(t) = m_h(t) \cos 2\pi f_c t + m(t) \sin 2\pi f_c t$
- $y(t) =$
 $m_h(t)(\sin(2\pi f_c t + \pi/2) - m(t) \cos(2\pi f_c t + \pi/2))$
- $y(t) =$
 $-[m(t) \cos(2\pi f_c t + \pi/2) - m_h(t) \sin(2\pi f_c t + \pi/2)]$
- Negative USSB signal for message signal $m(t)$ with phase shift of $\pi/2$.

CW Problem

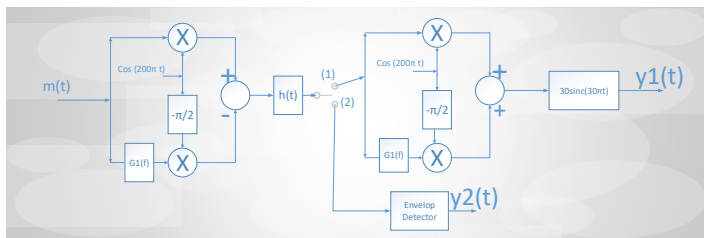
A SSB signal is generated by modulating an 800- KHz carrier by the signal

$$m(t) = \cos(2000\pi t) + 2 \sin(2000\pi t).$$

- 1 Determine the time-domain expression for the lower sideband of the SSB signal.

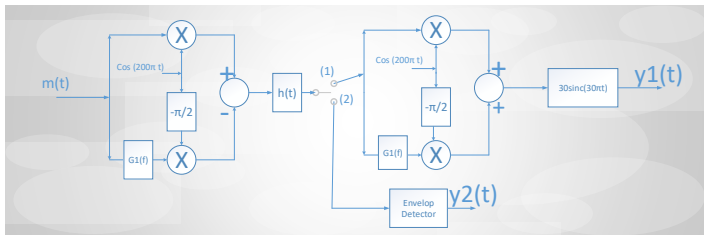
CW Problem

A message signal $m(t) = 20\text{sinc}(20\pi t)$ is transmitted over a channel $h(t) = \delta(t)$. The output of channel is either connected to switch (1) or switch (2). Take $G_1(f) = -j\text{sgn}(f)$ as the Hilbert filter.



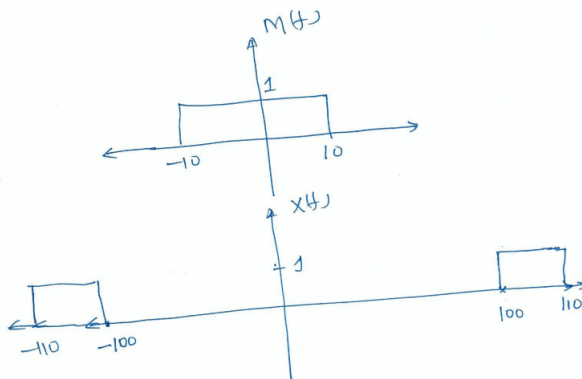
- 1 Find the bandwidth of transmitted signal.
- 2 Find the signal $y_1(t)$ if the switch is connected to (1).

Solution



Solution

- The signal at the input of channel is USSB
 $x(t) = m(t) \cos 200\pi t - m_h(t) \sin 200\pi t$.
Frequency spectrum is shown as follows:



BW=10 Hz.

Solution

- Signal after the channel output $z(t) = x(t)$.

Signal before LPF in the upper branch:

$$z_1(t) = [m(t) \cos 200\pi t - m_h(t) \sin 200\pi t] \cos 200\pi t.$$

In the lower branch:

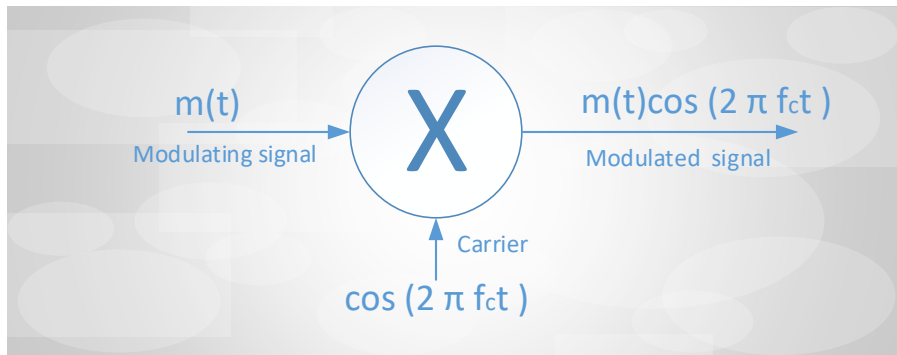
$$z_2(t) = [m_h(t) \cos 200\pi t + m(t) \sin 200\pi t] \sin 200\pi t.$$

Using trigonometric identities, and the effect of the LPF, the output $y(t) = m(t)$. Synchronous detector for USSB.

- The envelop of $z(t) = \sqrt{m^2(t) + m_h^2(t)}$.

Generation of AM Modulation

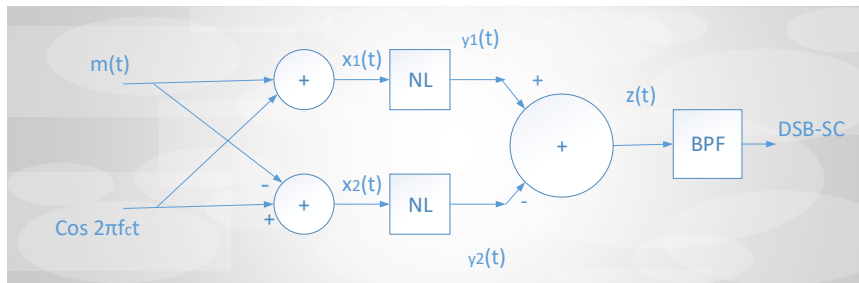
AM Generation: Multiplier



Earlier multipliers were difficult to implement !
Also, generation of a pure sinusoidal was difficult.

AM Generation: Nonlinear/Balanced Modulator

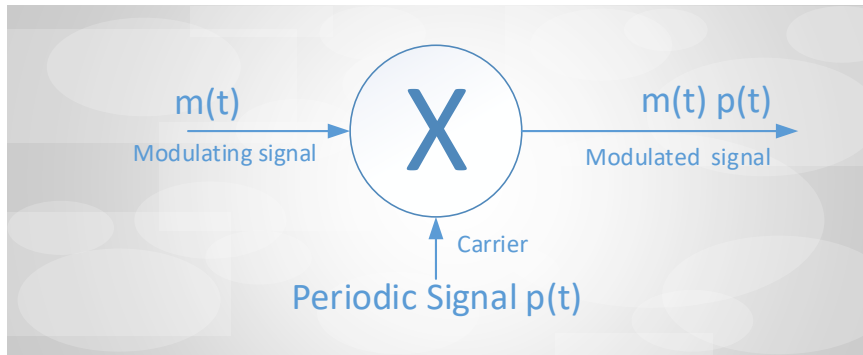
- Non-Linear device: $y(t) = ax(t) + bx^2(t)$



- Find the signal at the input of BPF.

AM Generation: Switching Modulator

- Multiplier input: replaced by a periodic signal of fundamental frequency f_c .



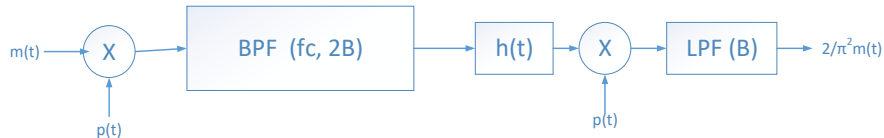
AM Generation: Switching Modulator

- Periodic signal



- Fourier series: $p(t) = \frac{1}{2} + \frac{2}{\pi} (\cos 2\pi f_c t - \frac{1}{3} \cos 3(2\pi f_c t) + \frac{1}{5} \cos 5(2\pi f_c t) - \dots)$

Transceiver

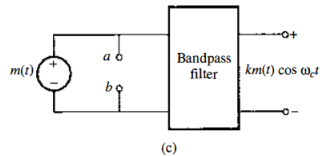
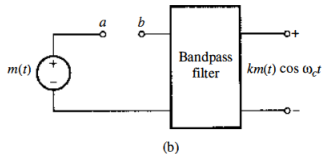
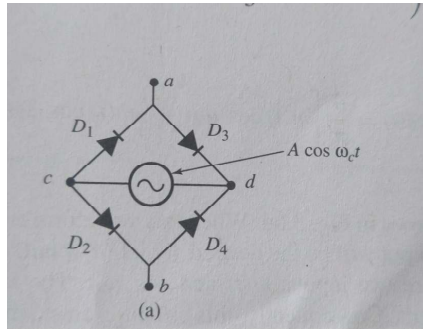


M2L15: Objectives of Today Lecture

- Circuits to generate AM Signals
- Demodulation Circuit



AM Generation: Diode-Bridge Modulator



Switching Action

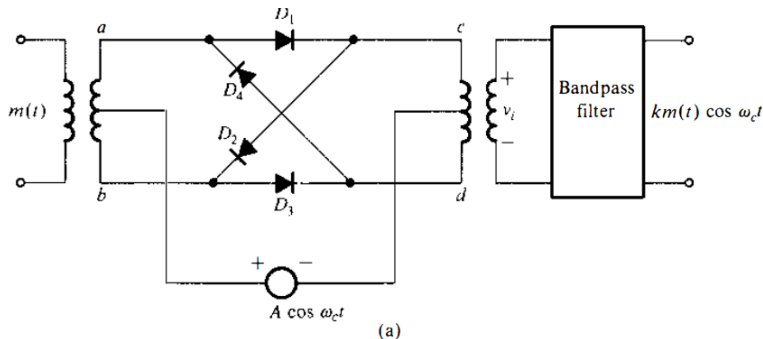
- Diodes act as switches controlled by carrier $c(t) = A \cos \omega_c t$.
- If $c(t) > 0$: All Diodes conduct.
- If $c(t) < 0$: All Diodes Open
- Switching effect:



- Fourier series:
$$p(t) = \frac{1}{2} + \frac{2}{\pi}(\cos 2\pi f_c t - \frac{1}{3} \cos 3(2\pi f_c t) + \frac{1}{5} \cos 5(2\pi f_c t) - \dots)$$
- $v_{ab} = m(t)p(t)$

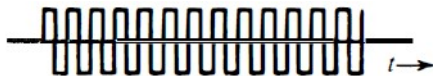
AM Generation: Ring Modulator

- Block diagram



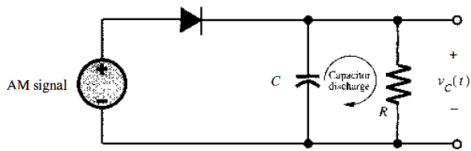
Switching Action

- Diodes act as switches controlled by carrier $c(t) = A \cos \omega_c t$.
- If $c(t) > 0$: D1, D3: ON, D2, D4: Off.
- If $c(t) < 0$: D1, D3: off, D2, D4: on.
- Switching effect:

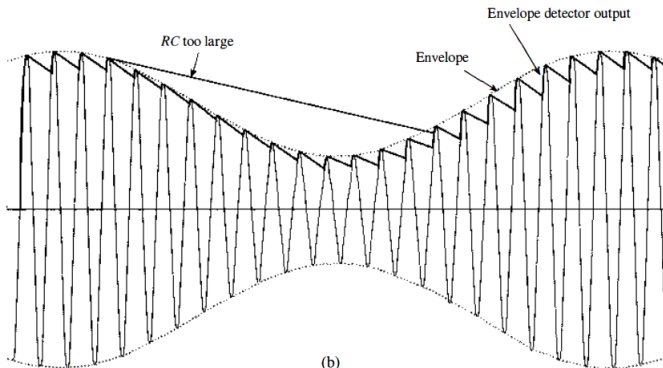


- Fourier series:
$$p(t) = \frac{4}{\pi} (\cos 2\pi f_c t - \frac{1}{3} \cos 3(2\pi f_c t) + \frac{1}{5} \cos 5(2\pi f_c t) - \dots)$$
- $v_i = m(t)p(t)$

Envelope Detection: Circuits



(a)



(b)

RC Envelope Detector: Operation

- **Forward bias (charging):** When $v_{in}(t) > v_C(t)$, the diode conducts.
 - Capacitor charges **quickly** to the input peak.
 - Charging path $\approx$ diode (very low R) $\rightarrow$ not governed by RC .
- **Reverse bias (discharging):** When $v_{in}(t) < v_C(t)$, the diode is OFF.
 - Capacitor discharges slowly through R .
 - Discharge follows $v_C(t) = V_{C0}e^{-t/RC}$.
 - This is the interval where envelope detection occurs.
- **Next conduction:** On the rising carrier, as soon as $v_{in}(t)$ exceeds the decayed $v_C(t)$, the diode conducts again and the capacitor **tops up** to the new envelope peak.

Effect of time constant RC on Envelop Detection

- If $RC \rightarrow \infty$: no capacitor discharge
- If $RC \rightarrow 0$: Output follows the input and not the envelop
- $RC < \frac{1}{B}$ to follow the envelop of the signal
- $RC \gg \frac{1}{f_c}$ not to follow the input signal
- Final condition: $\frac{1}{f_c} \ll RC < \frac{1}{B}$

CW Problem

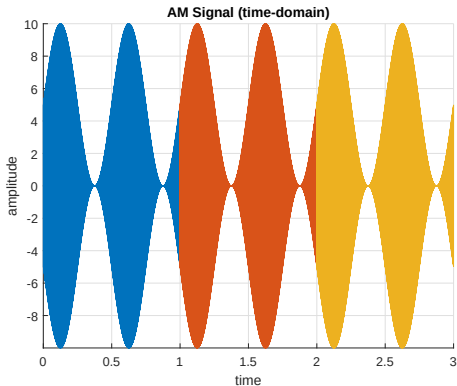
An AM signal $x(t) = (2 + \sin 20\pi t) \cos 200\pi t$. Find range of the capacitor for envelope detection using diode-RC circuit. Take $R = 10\Omega$.

Solution

- Condition for envelope detection: $\frac{1}{f_c} \ll RC < \frac{1}{f_m}$
- Using $f_c = 100$, $f_m = 10$, $R = 10$, we get $C < 0.01\text{F}$ and $C \gg 0.001\text{F}$.

Angle Modulation

Angle Modulation: Motivation



- Effect of noise?
- Power requirement?

Angle Modulation: Introduction

- Angle modulation: angle of the carrier wave is varied according to the message signal.
- Carrier signal: $A \cos(2\pi f_c t)$
- What is the angle? $\theta = 2\pi f_c t$
- What is the frequency? $f = \frac{1}{2\pi} \frac{d\theta}{dt} = f_c$.

Instantaneous Phase and Instantaneous Frequency

- $x(t) = A \cos(2\pi f_c t + \phi(t))$
- Instantaneous phase: $\phi(t)$:
- What is the instantaneous frequency of $x(t)$?
- $f_i(t) = f_c + \frac{1}{2\pi} \frac{d\phi(t)}{dt}$.
- What is frequency deviation?
- $\Delta f(t) = f_i(t) - f_c = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$.

Instantaneous Phase Deviation and Instantaneous Frequency Deviation

- Un-modulated carrier: $A \cos(2\pi f_c t)$
- Angle Modulated: $x(t) = A \cos(2\pi f_c t + \phi(t))$
- Instantaneous phase deviation: $\Delta\phi(t) = \phi(t)$:
- What is instantaneous frequency deviation?
- $\Delta f(t) = f_i(t) - f_c = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$.

Angle Modulation: Phase Modulation

- PM: $\phi(t) \propto m(t)$
- $\phi(t) = k_p m(t)$
- k_p : phase sensitivity (radians per volt)
- $x_{PM}(t) = A \cos(2\pi f_c t + k_p m(t))$

Angle Modulation: Frequency Modulation

- FM: $\Delta f(t) (= f_i(t) - f_c) \propto m(t)$
- $f_i(t) = f_c + k_f m(t)$
- k_f : frequency sensitivity (Hz per volt).
- $x_{FM}(t) = A \cos(2\pi f_c t + \phi(t))$
- $f_i(t) - f_c = k_f m(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$
- $\phi(t) = 2\pi k_f \int_{-\infty}^t m(\tau) d\tau$
- $x_{FM}(t) = A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$

M2L16: Objectives of Today Lecture

- Angle Modulation (FM and PM)



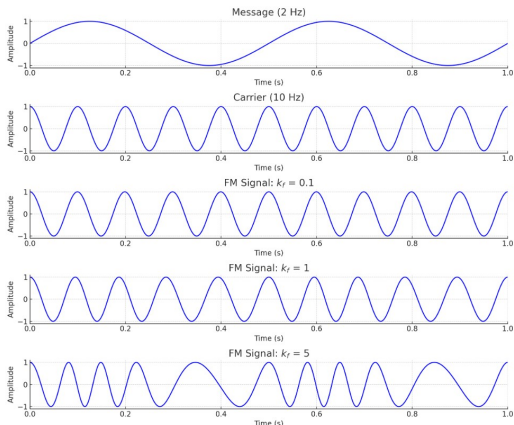
Phase Modulation: Summary

- $x_{PM}(t) = A \cos(2\pi f_c t + k_p m(t))$
- Unit of $k_p m(t)$: radians
- Unit of k_p : radians/unit of $m(t)$
- $f_i(t) = f_c(t) + \frac{1}{2\pi} k_p \frac{dm(t)}{dt}$.
- Maximum phase deviation $\Delta\phi = k_p m_p$
- m_p is maximum amplitude of $m(t)$ as defined for AM.
- Maximum frequency deviation :
$$\Delta f = \frac{1}{2\pi} k_p \max \frac{dm(t)}{dt}$$

Frequency Modulation: Summary

- $x_{FM}(t) = A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$
- Unit of $2\pi k_f \int_{-\infty}^t m(\tau) d\tau$: radians
- Unit of $2\pi k_f$: radians/unit of $m(t)$
- Unit of k_f : Hz/unit of $m(t)$
- $f_i(t) = f_c(t) + k_f m(t)$.
- Maximum frequency deviation
 $\Delta_f = k_f \max m(t) = k_f m_p$.
- Maximum (or peak) phase deviation
 $\Delta\phi = 2\pi k_f \max \int_{-\infty}^t m(\tau) d\tau$

Angle Modulation: Sketch



Angle Modulation: Example

$$x(t) = A \cos(200\pi t + 2 \cos 20\pi t)$$

Angle Modulation: Modulation Index β

- Maximum frequency deviation: $\Delta f = \max \Delta f(t)$
- Maximum frequency of message signal: f_m
- $\beta = \frac{\Delta f}{f_m}$

CW Problem

An angle modulated signal has the form

$$x(t) = 100 \cos(2\pi 10^7 t + 4 \sin 2000\pi t)$$

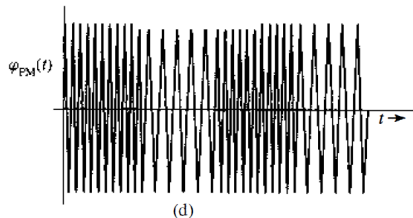
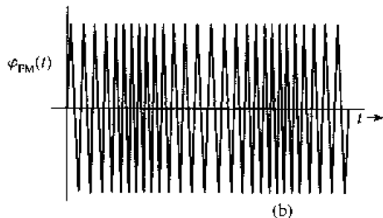
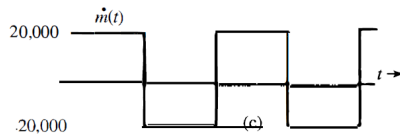
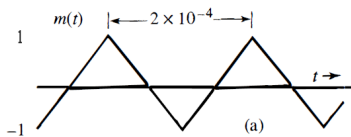
- Determine average transmitted power.
- Determine the peak-phase deviation.
- Determine the peak-frequency deviation.
- Is this an FM/PM?

Solution

- $P = A^2/2 = 100^2/2 = 5000.$
- Maximum phase deviation
 $\Delta\phi_{\max} = \max 4 \sin 2000\pi t = 4.$
- Instantaneous frequency
 $f_i = f_c + \frac{1}{2\pi} \frac{d}{dt} \theta(t) = f_c + 4000 \cos(2000\pi t)$
- Maximum frequency deviation $\Delta f_{\max} = 4000.$
- The angle modulated can be both PM or FM
- PM: $k_p = 4, m(t) = \sin(2000\pi t).$
- FM signal $k_f = 4000, m(t) = \cos(2000\pi t)$

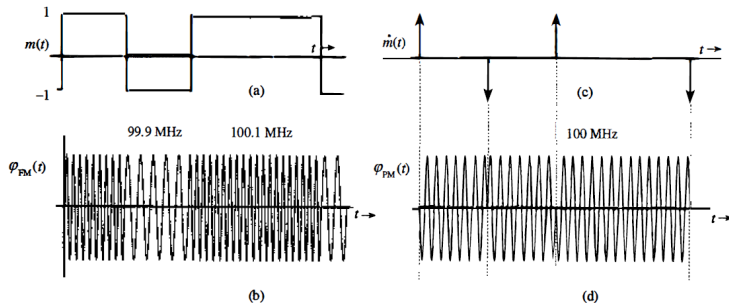
Angle Modulation: Sketch

- $k_f = 2\pi \times 10^5$; $k_p = \pi/2$, $f_c = 100$ MHz
- FM $f_i(t) = 10^8 + 10^5 m(t)$
- PM $f_i(t) = f_c + \frac{k_p}{2\pi} \frac{dm(t)}{dt} = 10^8 + 5 \frac{dm(t)}{dt}$



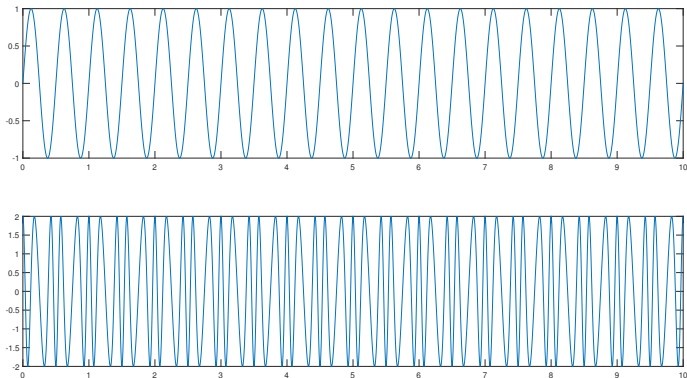
Angle Modulation: Sketch

- $k_f = 2\pi \times 10^5$; $k_p = \pi/2$, $f_c = 100$ MHz
- FM $f_i(t) = 10^8 + 10^5 m(t)$
- PM $f_i(t) = f_c + \frac{k_p}{2\pi} \frac{dm(t)}{dt} = 10^8 + \frac{1}{4} \frac{dm(t)}{dt}$



Angle Modulation: Sketch

- $x(t) = 2 \cos(10\pi t + 5 \sin(4\pi t))$



M2L17: Objectives of Today Lecture

- BW of Angle Modulation (FM and PM)
- Generation of FM Signals



CW Problem

Find maximum frequency deviation:

- $x(t) = 5 \cos(10\pi t + t)$

FM Modulation: Bandwidth

- $x_{FM}(t) = A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$
- $x_{FM}(t) = A \cos(\omega_c t + k_f \int_{-\infty}^t m(\tau) d\tau)$
- Let us assume $a(t) = \int_{-\infty}^t m(\tau) d\tau$
- $x_{FM}(t) = A \cos(\omega_c t + k_f a(t))$
- $x_{FM}(t) = \Re(A e^{j[\omega_c t + k_f a(t)]})$
- $x_{FM}(t) = A \Re[e^{jk_f a(t)} e^{j\omega_c t}]$
- $x_{FM}(t) =$
 $A \Re \left[\left[1 + jk_f a(t) - \frac{1}{2!} k_f^2 a^2(t) + \dots + j^n \frac{k_f^n}{n!} a^n(t) + \dots \right] e^{j\omega_c t} \right]$
-

$$x_{FM}(t) = A [\cos(\omega_c t) - k_f a(t) \sin \omega_c t - \frac{1}{2!} k_f^2 a^2(t) \cos \omega_c t + \dots]$$

FM Modulation: Bandwidth

- The modulated wave consists of an modulated carrier plus various amplitude modulated terms such as $a(t) \sin(\omega_c t)$, $a^2(t) \cos^2(\omega_c t)$and so on.
- Bandwidth of $a(t) : B$
- Bandwidth of $a^2(t) : 2B$
- Bandwidth of $a^n(t) : nB$
- Thus, spectrum of $x_{FM(t)}$ consists of an unmodulated carrier plus spectra of DSB-SC signals with messages $a(t)$, $a^2(t)$,.... $a^n(t)$,
- Even $m(t)$ is band-limited, $x_{FM(t)}$ is not band-limited
- Bandwidth of FM = ∞

FM Modulation: Bandwidth

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- Thus, spectrum of $x_{FM(t)}$ consists of an unmodulated carrier plus spectra of DSB-SC signals with messages $a(t)$, $a^2(t)$,.... $a^n(t)$,
- Even $m(t)$ is band-limited, $x_{FM(t)}$ is not band-limited
- Bandwidth of FM = ∞

FM Modulation: Effective Bandwidth

- For practical message signal $m(t)$, $\frac{k_f^n a^n(t)}{n!} \approx 0$, as n increases.
- Most of the modulated signal-power resides in a finite bandwidth.

Minimum f_c

$$f_c - nB \geq 0$$

Phase Modulation: Bandwidth

- $x_{PM}(t) = A \cos(\omega_c t + k_f m(t))$
- Similar to the FM (substitute $a(t) = m(t)$):

$$x_{PM}(t) = A \left[\cos(\omega_c t) - k_f m(t) \sin \omega_c t - \frac{1}{2!} k_f^2 m^2(t) \cos \omega_c t + \dots \right] \quad (1)$$

- Since BW of $m(t)$ and $a(t)$ is same, all conclusions drawn for FM, holds true for PM too, and in general for any angle modulated scheme.

Bandwidth of Angle Modulation: General Signals $m(t)$

- Carson's rule for effective BW (98% of power)
- $B_{\text{angle modulation}} = 2(\Delta f + B)$ (Ideally, we should use $\approx$ sign)
- Δf : Frequency deviation
- B : Message bandwidth
- If higher frequency content is given: f_m .
- $B_{\text{angle modulation}} = 2(\Delta f + f_m)$
- For truly wide-band $\Delta f \gg B$,
 $B_{\text{angle modulation}} \approx 2\Delta f$
- For narrow-band $\Delta f \ll B$,
 $B_{\text{angle modulation}} \approx 2B$

FM Modulation: Narrow band Approximation

- NBFM
- Narrow is related to bandwidth of the modulated signal and not for the message bandwidth B .
- When $k_f \ll 1$ or $\beta_f \ll 1$ such that $|k_f a(t)| \ll 1$
- From series expansion:

$$x_{NBFM}(t) \approx A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

$$x_{NBFM}(t) \approx A \left[\cos \omega_c t - k_f \sin \omega_c t \int_{-\infty}^t m(\tau) d\tau \right]$$

- Bandwidth of $x_{NBFM}(t)$ is $2B$

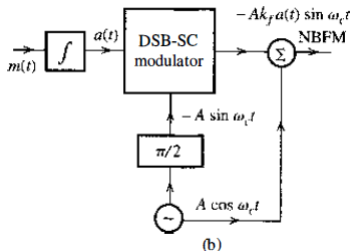
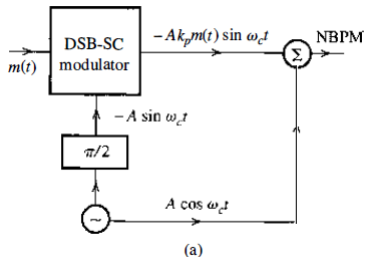
Phase Modulation: Bandwidth

- When $k_p \ll 1$ or $\beta_p \ll 1$ such that $|k_p m(t)| \ll 1$
- NBPM

$$x_{NBPM}(t) \approx A [\cos \omega_c t - k_p m(t) \sin \omega_c t]$$

- Comparing, NBPM is very similar to the AM.

Narrow-band Implementation



Draw FM Transmitter

Use of BPF?

Angle Modulation: Single Tone

- $m(t) = A_m \cos(2\pi f_m t)$
- $x_{FM}(t) = A_c \cos(2\pi f_c t + \frac{k_f A_m}{f_m} \sin(2\pi f_m t))$
- $x_{FM}(t) = A_c \cos(2\pi f_c t + \beta_f \sin(2\pi f_m t))$

FM BW: Single-Tone Modulation

- $m(t) = A \cos(2\pi f_m t)$
- $x_{FM}(t) = A \cos(2\pi f_c t + \beta_f \sin(2\pi f_m t))$
- $x_{FM}(t) = A \operatorname{Re}(e^{j2\pi f_c t} e^{j\beta_f \sin(2\pi f_m t)})$
- Expansion of $e^{j\beta_f \sin(2\pi f_m t)}$ with period $T_0 = 1/f_m$
- General formula Fourier Series:
$$g(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn2\pi f_0 t}$$
- where $c_n = \frac{1}{T_0} \int_{T_0} g(t) e^{-jn2\pi f_0 t} dt$
- Fourier series exponential coefficients for our equation:
- $c_n = f_m \int_0^{1/f_m} e^{j\beta_f \sin(2\pi f_m t)} e^{-jn2\pi f_m t} dt$

FM Modulation: Single-Tone Modulation

- $c_n = f_m \int_0^{1/f_m} e^{j\beta_f \sin(2\pi f_m t)} e^{-jn2\pi f_m t} dt$
- Substitute $u = 2\pi f_m t$
- $c_n = \frac{1}{2\pi} \int_0^{2\pi} e^{j(\beta_f \sin u - nu)} du$
- $c_n = J_n(\beta_f)$
- $J_n(\beta_f)$: Bessel function of the first kind of order n .

FM Modulation: Single-Tone Modulation

- Fourier series:

$$e^{j\beta_f \sin(2\pi f_m t)} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t}$$

- $x_{FM}(t) = A \operatorname{Re}(e^{j2\pi f_c t} e^{j\beta_f \sin(2\pi f_m t)})$
- $x_{FM}(t) = A \operatorname{Re}(e^{j2\pi f_c t} \sum_{n=-\infty}^{\infty} J_n(\beta_f) e^{j2\pi n f_m t})$
- $x_{FM}(t) = \sum_{n=-\infty}^{\infty} A J_n(\beta_f) \cos(2\pi(f_c + n f_m)t)$
- Same observation: BW is infinite even for a single-tone modulation.

M2L18: Objectives of Today Lecture

- BW of Angle Modulation (FM and PM)
- Generation of FM Signals
- Demodulation



Bandwidth of Angle Modulation: General Signals $m(t)$

- Carson's rule for effective BW (98% of power)
- $B_{\text{angle modulation}} = 2(\Delta f + B)$ (Ideally, we should use $\approx$ sign)
- Δf : Frequency deviation
- B : Message bandwidth
- If higher frequency content is given: f_m .
- $B_{\text{angle modulation}} = 2(\Delta f + f_m)$
- For truly wide-band $\Delta f \gg B$,
 $B_{\text{angle modulation}} \approx 2\Delta f$
- For narrow-band $\Delta f \ll B$,
 $B_{\text{angle modulation}} \approx 2B$

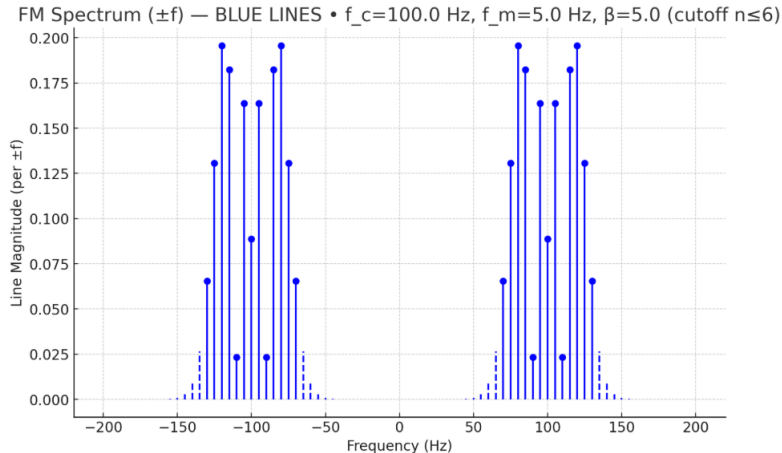
Single-Tone FM

- $x_{FM}(t) = \sum_{n=-\infty}^{\infty} A J_n(\beta_f) \cos(2\pi(f_c + n f_m)t)$
- The Bessel function of the first kind:

$$J_n(\beta) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + n + 1)} \left(\frac{\beta}{2}\right)^{2m+n}.$$

- Carson's rule: $J_n(\beta)$ becomes negligible when $n > \beta + 1$
- $B_{FM} = 2(\beta + 1)f_m = 2(\Delta_f + f_m)$

Single-Tone FM



CW Problem

An FM modulator is followed by an ideal bandpass filter having a center frequency of 500 Hz and a bandwidth of 70 Hz. The gain of the filter is 1 in the passband. The unmodulated carrier is given by $10 \cos(1000\pi t)$, and the message signal is $m(t) = 10 \cos(20\pi t)$. The transmitter frequency sensitivity is $k_f = 8 \text{ Hz/Volt}$.

- 1 Determine the peak frequency deviation in Hertz.
- 2 Determine the peak phase deviation in radians.
- 3 Determine the modulation index.
- 4 Determine the power at the filter input
- 5 Determine the power at the filter output

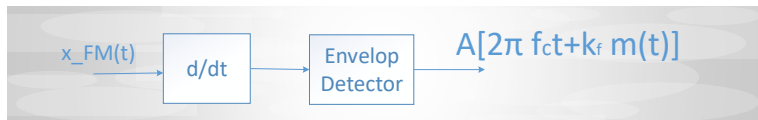
Solution

- Assuming signals at initial condition $t < 0$ is zero,
- FM equation:
$$x_{FM}(t) = 10 \cos(1000\pi t + 2\pi k_f \int_0^t m(\tau) d\tau) = 10 \cos(1000\pi t + 8 \sin(20\pi t)).$$
- The instantaneous frequency deviation
$$f_i(t) = \frac{1}{2\pi} \frac{d\theta_i(t)}{dt} = 500 + 80 \sin(10\pi t).$$
 Thus, maximum frequency deviation $\Delta f = 80$ Hz.
- $x_{FM}(t) = 10 \cos(1000\pi t + 8 \sin(20\pi t)).$
- Peak phase deviation : 8 radians.
- $\beta_f = \frac{\Delta f}{f_m} = 80/10 = 8$

Solution

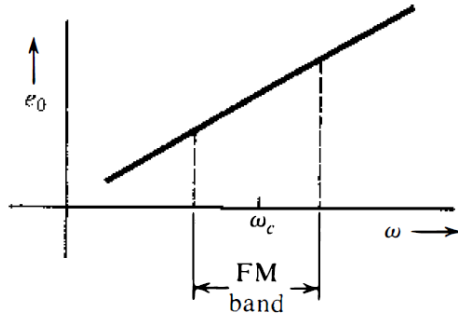
- Power of $x_{FM}(t) = 10 \cos(1000\pi t + 8 \sin(20\pi t))$ is $10^2/2 = 50$ Watts.
- $x_{FM}(t) = 10 \cos(1000\pi t + 8 \sin(20\pi t)) = 10 \sum_{n=-\infty}^{\infty} J_n(8) \cos(2\pi[500 + n10]t)$.
- The bandpass filter will pass only signals between 465 Hz and 535 Hz.
- Thus, the output signal will be $10J_0(8) \cos(2\pi 500t) + 10J_1(8) \cos(2\pi 510t) + 10J_{-1}(8) \cos(2\pi 490t) + 10J_2(8) \cos(2\pi 520t) + 10J_{-2}(8) \cos(2\pi 480t) + 10J_3(8) \cos(2\pi 530t) + 10J_{-3}(8) \cos(2\pi 470t)$.
- The output power will be $50|J_0(8)|^2 + 2 \times 50|J_2(8)|^2 + 2 \times 50|J_1(8)|^2 + 2 \times 50|J_3(8)|^2$.

Direct Method for FM Demodulation: Discriminator



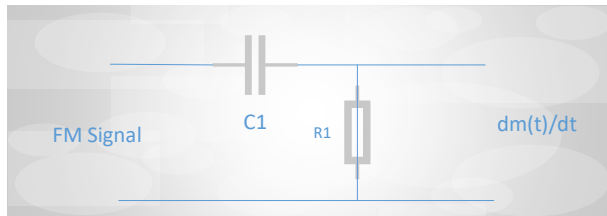
Note: 2π is missing in k_f

Slope Detector



Practical Differentiation

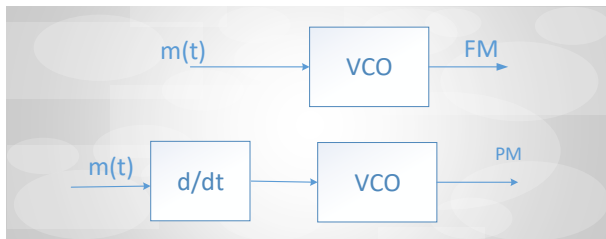
- An example of differentiation transfer function $H(f) = j2\pi f$
- RC high pass- filter



- $H(f) = \frac{j2\pi fRC}{1+j2\pi fRC} \approx j2\pi fRC$ if $2\pi fRC \ll 1$
- It is called as the slope detector because the operation is on the slope of $H(f)$.

FM Generation: Direct Method

- The angle modulators are non-linear systems and time-varying, in general.
- Direct Generation
- Voltage Controlled Oscillator: the frequency is controlled by an external voltage.
- If no applied $m(t)$, $f_i(t) = f_c$.



Contd.

- Oscillator frequency : $f_c = \frac{1}{2\pi\sqrt{L_0C_0}}$ without control voltage
- If the capacitance is varied according to the modulating signal $m(t)$: $C = C_0 - km(t)$
- $f_i(t) = \frac{1}{2\pi\sqrt{L_0C_0}\sqrt{[1 - \frac{km(t)}{C_0}]}} \approx \frac{1}{2\pi\sqrt{L_0C_0}}[1 + \frac{km(t)}{2C_0}]$
- We have used $\frac{km(t)}{C_0} \ll 1$.
- $f_i(t) = f_c[1 + \frac{km(t)}{2C_0}]$
- $f_i(t) = f_c + \frac{kf_c}{2C_0}m(t)$
- $f_i(t) = f_c + k_fm(t)$
- Where $k_f = \frac{kf_c}{2C_0}$

M2L19: Objectives of Today Lecture

- Armstrong Method for Generation of FM Signals
- Superhetrodyne Receiver



CW Problem

Estimate the bandwidth of FM signal for $m(t) = e^{-t^2}$,
 $f_c = 10^4\text{Hz}$, $k_f = 3000$.

Solution

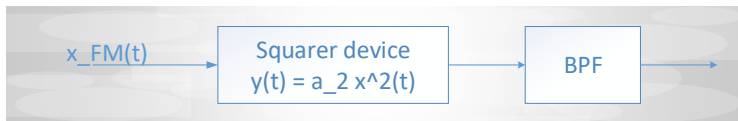
- $m(t) = e^{-t^2}$
- $M(f) = \sqrt{\pi}e^{-\pi^2 f^2}$
- BW is very small.
- $\Delta_f = 3\text{KHz}$
- FM BW = 6 KHz.

FM Generation: Indirect Method (Armstrong Method)

- First generate NBFM $\Delta_f \ll 1$
- Pass the NBFM to frequency multipliers to get WBFM with an increased Δ_f but with an increased f_c

Frequency Multiplier

- Frequency multiplier contains a non-linear device and a BPF.

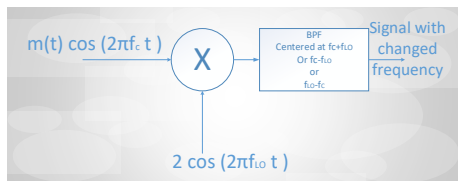


n-order Frequency Multiplier

- $y(t) = a_0 + a_1x(t) + a_2x^2(t) + ...a_nx^n(t)$
- $x(t) = A \cos[2\pi f_c t + 2\pi k_f \int m(\tau) d\tau]$
- $f_i(t)$ is increased by n -times.

Frequency Mixer/Converter

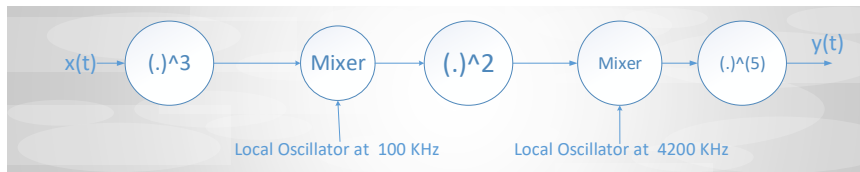
- Frequency mixer: Multiplier plus BPF



- Find the signal input to the BPF
- $m(t)[\cos 2\pi(f_c + f_{LO})t + \cos 2\pi(f_c - f_{LO})t]$
- Since $\cos$ is even function:
- $m(t)[\cos 2\pi(f_c + f_{LO})t + \cos 2\pi(f_{LO} - f_c)t]$
- Output frequency will contain either $f_c + f_{LO}$ or $f_c - f_{LO}$ or $f_{LO} - f_c$ depending on the center frequency of BPF.

Armstrong Method

- FM signal $x(t)$ with $m(t) = \text{sinc}(2\pi t)$;
 $f_c = 1000\text{KHz}$, $k_f = 1$. Find carrier and frequency deviation at the output. Assume center frequency of the BPF as the sum of two input frequencies.



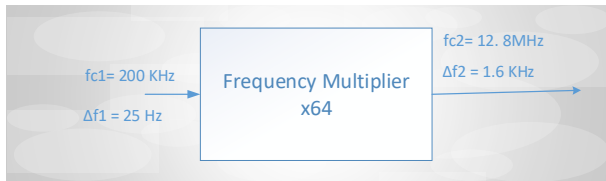
Armstrong Indirect FM Generation: Example

- NBFM $f_{c1} = 200$ KHz, $\Delta_{f1} = 25$ Hz.
- Desired output: Carrier frequency 91.2 MHz, $\Delta_f = 75$ KHz.
- How-many multiplications : $75000/25 = 3000$
- A slight different in Δ_f is tolerable, but not in f_c .

Armstrong Indirect FM Generation: Example

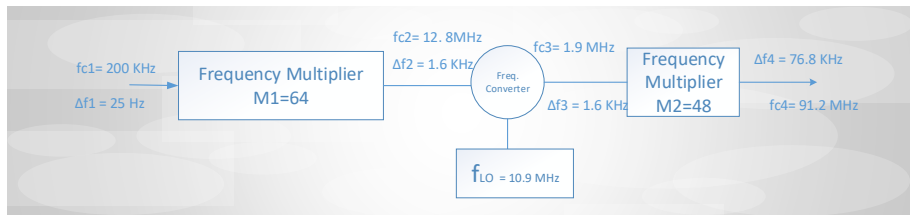
- A single multiplier stage: $f_{cfinal} = f_{c1} \times 3000 = 600 \text{ MHz}$.
- Mixing with a high frequency oscillator of 508.8 MHz may work. What if such high frequency oscillator is not available.
- We need to have multiple stages: each stage has a frequency converter to control the carrier frequency.
- Let us have two stage: $M_1 = 64$, $M_2 = 48$: total multiplications: $64 \times 48 = 3072$.
- First stage : $f_{c2} = M_1 \times f_{c1} = 64 \times 200 \text{ KHz} = 12.8 \text{ MHz}$.
- $\Delta f_2 = 64 \times 25 \text{ Hz} = 1.6 \text{ KHz}$

Armstrong Indirect FM Generation: Example



- $f_{c3} = \frac{91.2 \text{ MHz}}{48} = 1.9 \text{ MHz}$
- What is oscillator frequency to maintain $f_{c3} = 1.9 \text{ MHz}$ for the incoming $f_{c2} = 12.8 \text{ MHz}$
- Choice 1: $f_{LO} = f_{c2} + f_{c3} = 14.7 \text{ MHz}$
- Choice 2: $f_{LO} = f_{c2} - f_{c3} = 10.9 \text{ MHz}$
- Choice 3: $f_{LO} = f_{c3} - f_{c2} = -10.9 \text{ MHz}$
- If there is no restriction on LO: chose any LO with positive frequency

Armstrong Indirect FM Generation: Example



Problem

Design (only the block diagram) an Armstrong indirect FM modulator to generate an FM carrier with a carrier frequency of 96 MHz and $\Delta f = 20$ kHz. A narrowband FM generator with $f_c = 200$ kHz and adjustable Δf in the range of 9 to 10 Hz is available. The stock room also has an oscillator with adjustable frequency in the range of 9 to 10 MHz. There is a bandpass filter with any center frequency, and only frequency doublers are available.

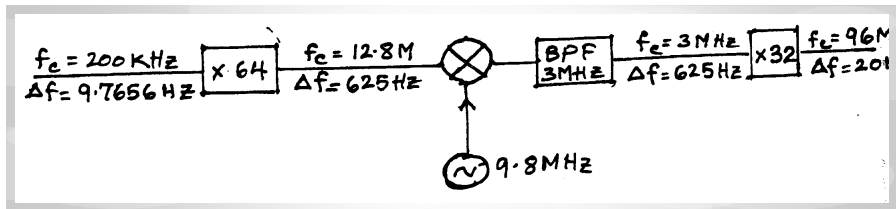
Solution

- NBFM generates: $f_{c1} = 200$ KHz, and $\Delta_{f1} = 9$ to 10 Hz.
- The final WBFM : $f_c = 96$ MHz and $\Delta_f = 20$ KHz.
- Total number of frequency multiplier needed: $\frac{\Delta_f}{\Delta_{f1}} = 2000$ to 2200.
- Since only doublers are available: closest n : $2^n = 2000$ 2222
- $n = 11$, thus $n^{11} = 2048$.
- Fix $\Delta_{f1} = \frac{\Delta_f}{2^{11}} = 9.7656$ Hz.
- If this flexibility (initial frequency deviation of NBFM) was not possible, there will be a slight difference in deviation but f_c should match exactly.
- One stage will not work: carrier frequency become $200\text{kHz} \times 2048 = 409.6$ MHz.
- We have to add frequency mixers to control carrier frequency.

Solution

- Let us make two stages $M_1 = 2^{n_1}$ and $M_2 = 2^{n_2}$
- $f_{c2} = 2^{n_1} f_{c1}$ and $f_c = 2^{n_2} f_{c3}$, where f_{c3} is after mixer.
- To find f_{c3} and f_{LO}
- $f_{c3} = f_{c2} - f_{LO}$, $f_{c3} = f_{c2} + f_{LO}$, and $f_{c3} = f_{LO} - f_{c2}$.
- Each should be tested $9MHz < f_{LO} < 10MHz$.
- $M_1 = 64$, $M_2 = 32$, $f_{LO} = 9.8MHz$

Block Diagram



Radio Transmissions and Reception

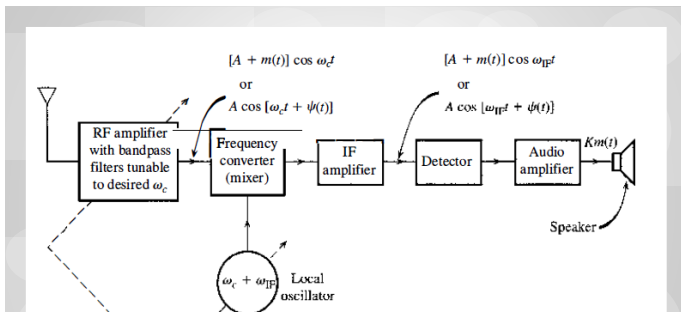
- AM radio: Frequency band allocation 535 KHz-1605 KHz
- AM Radio: Carrier spacing 10 KHz
- Calculate number of AM radio station
- 107
- FM radio: Frequency band allocation 88 MHz-108 MHz
- FM Radio: Carrier spacing 200 KHz,
 $\Delta_f = 75 KHz$
- Calculate number of FM radio stations
- 100

Bandpass filtering for Radio Station Tuning

- In AM radio: the carriers are only 10 KHz apart.
- In FM radio: the carriers are only 200 KHz apart.
- Quality factor of BPF: $Q = \frac{f_c}{B}$
- A bandpass filter at the RF may not provide enough selectivity for carriers separated by 10 KHz or even 200 KHz.

Superhetrodyne Receiver

- Intermediate frequency (IF) amplifier: 455 KHz (AM), BW 10 KHz, 10.7 MHz (FM), BW 200 KHz, 38 MHz(TV), BW 6 MHz.
- Frequency of oscillator fixed : $f_{LO} = f_c + f_{IF}$ or $f_{LO} = f_c - f_{IF}$.
- Frequency at the output of IF should be: f_{IF} .



Superhetrodyne Receiver: Role of IF amplifier

- Rejects other carrier frequencies or radio stations.
- It provides a constant frequency to the detector and subsequent stages of receiver for any input carrier signal at the RF: superhetrodyne
- Picking f_{LO} higher than f_c : easier to design an oscillator which is tunable for a smaller range.
- $f_{LO} = f_c + f_{IF}$ ranges from 1005 KHz to 2060 KHz (ratio approximately 2)
- If $f_{LO} = f_c - f_{IF}$ ranges from 95 KHz to 1150 KHz (ratio approximately 12)

Superhetrodyne Receiver: Image frequency

Stations apart from $2f_{IF}$

Superhetrodyne Receiver: Role of RF filter at the Front End

- The RF bandpass filter tuned at f_c provide good selectivity for image frequency at $f_c + 2f_{IF}$.
- RF filter is used to suppress the image station.

Problem: TB 4.6-3

Consider a superheterodyne FM receiver designed to receive the frequency band of 1 to 30 MHz with an IF frequency 8 MHz. What is the range of frequencies generated by the local oscillator for this receiver? An incoming signal with a carrier frequency of 10 MHz is received at the 10 MHz setting. At this setting of the receiver, we also get interference from a signal with some other carrier frequency if the receiver RF-stage bandpass filter has poor selectivity. What is the carrier frequency of the interfering signal?

Solution

- Carrier frequency range : 1 to 30 MHZ
- IF=8 MHZ
- LO= 9 to 38
- $f_c = 10$ MHZ, LO= 18 MHZ
- Another carrier image frequency $f_c = f_c + 2f_{IF}$
=26 MHZ

TB Problem

A transmitter transmits an AM signal with a carrier frequency of 1530 kHz. When an inexpensive radio receiver (which has a poor selectivity in its RF-stage bandpass filter) is tuned to 1530 kHz, the signal is heard loud and clear. This same signal is also heard (not as well) at another dial setting. State, with reasons, at what frequency you will hear this station. The IF frequency is 455 kHz.

620 KHz, Image $2 \times 455 = 910$ KHz.

CW Problem

A superheterodyne FM receiver operates in the frequency range of $88 - 108$ MHz. The IF and local oscillator frequencies are chosen such that $f_{IF} < f_{LO}$. We require that the Image frequency f'_c fall outside of the $88 - 108$ MHz region. Determine the minimum required f_{IF} and the range of variations in f_{LO} ?

Solution

- In FM carrier frequency range 88 – 108 MHz.
- Standards for FM radio, local oscillator is at $f_c + f_{IF}$.
- The RF BPF can reject the image frequency at $2f_{IF} + f_c$
- We select IF bandwidth such that image occurs outside the range 88 – 108 MHz.
- We want image for lowest carrier frequency $f_c = 88$ MHz to fall outside the range 88 – 108 MHz.
- $88 + 2f_{IF} > 108$, which gives $f_{IF} > 10$ MHz.
- Thus, if $IF > 10$ MHz, the image frequency always lie outside the FM range.

FM Discriminator

- 3.36 An FM discriminator is shown in Figure P-3.36. The envelope detector is assumed to be ideal and has an infinite input impedance. Select the values for L and C if the discriminator is to be used to demodulate an FM signal with a carrier $f_c = 80$ MHz and a peak-frequency deviation of 6 MHz.

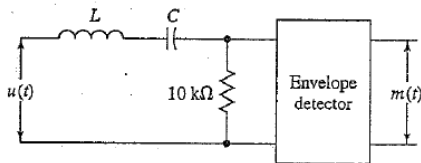


Figure P-3.36

Ans: $L = 14.98\text{mH}$ and $C = 0.018013\text{ pF}$.